

Problems in Three Dimensions

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In this chapter we discuss the structure of the Schrödinger equation for a particle moving in three dimensions. General properties are developed through examination of the free-particle problem in Cartesian and spherical coordinates. Separation of variables in spherical coordinates yields product solutions for the free-particle problem comprised of spherical harmonics and spherical Bessel functions. Solution to the corresponding radial wave equation for the hydrogen atom gives Laguerre polynomials. Application is also directed toward the motion of a charged particle in a magnetic field. An elementary description of the theory of radiation from atoms and the formulation of selection rules are given. The chapter concludes with a description of the Thomas–Fermi model important to atomic physics.

10.1 ■ THE FREE PARTICLE IN CARTESIAN COORDINATES

We again recall that the linear momentum operator $\hat{\mathbf{p}}$ is given by

$$\hat{\mathbf{p}} = -i\hbar\nabla \quad (10.1)$$

Inserting this form into the Hamiltonian for a free particle of mass m moving in three dimensions gives

$$\hat{H} = \frac{\hat{p}^2}{2m} = -\frac{\hbar^2}{2m}\nabla^2 \quad (10.2)$$

It follows that the time-independent Schrödinger equation for this same particle appears as

$$\hat{H}\varphi = -\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \varphi = E\varphi \quad (10.3)$$

or, alternatively,

$$\begin{aligned} \nabla^2 \varphi &= -k^2 \varphi \\ E &= \frac{\hbar^2 k^2}{2m} \end{aligned} \quad (10.4)$$

Separating variables

$$\varphi \equiv X(x)Y(y)Z(z) \quad (10.5)$$

permits (10.4) to be rewritten as

$$\begin{aligned} \frac{X_{xx}}{X} + \frac{Y_{yy}}{Y} + \frac{Z_{zz}}{Z} &= -k^2 \\ -\frac{X_{xx}}{X} &= k^2 + \left(\frac{Y_{yy}}{Y} + \frac{Z_{zz}}{Z} \right) \equiv k_x^2 \end{aligned} \quad (10.6)$$

In the last equation, the left-hand side is a function only of x , while the middle term is a function only of y and z . The only way for the equality to hold for all (x, y, z) is for both terms to be equal to the same constant. Labeling this constant k_x^2 gives the equation

$$X_{xx} + k_x^2 X = 0 \quad (10.7)$$

which has a solution¹

$$X = A' e^{ik_x x} \quad (10.8)$$

In similar manner we obtain

$$Y = B' e^{ik_y y}, \quad Z = C' e^{ik_z z} \quad (10.9)$$

where

$$k^2 = k_x^2 + k_y^2 + k_z^2 \quad (10.10)$$

Combining all three factors X , Y , and Z gives the solution

$$\varphi = A' B' C' \exp[i(k_x x + k_y y + k_z z)] = \varphi_{\mathbf{k}} = A e^{i\mathbf{k} \cdot \mathbf{r}} \quad (10.11)$$

¹Here we consider only the forward propagating wave.

The wave vector $\mathbf{k}$ and position vector $\mathbf{r}$ have components

$$\begin{aligned}\mathbf{k} &= (k_x, k_y, k_z) \\ \mathbf{r} &= (x, y, z)\end{aligned}\tag{10.12}$$

The function $\varphi_{\mathbf{k}}$ so obtained is an eigenfunction of $\hat{H}$ (10.2) with the eigenvalue

$$E_k = \frac{\hbar^2 k^2}{2m}\tag{10.13}$$

Plane Waves

The corresponding solution to the time-dependent Schrödinger equation (3.52) appears as

$$\begin{aligned}\psi_{\mathbf{k}}(\mathbf{r}, t) &= A e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \\ \hbar \omega &= E_k\end{aligned}\tag{10.14}$$

This solution represents a propagating plane wave. At any instant of time, $\psi_{\mathbf{k}}(\mathbf{r}, t)$ is constant on the surfaces $\mathbf{k} \cdot \mathbf{r} = \text{constant}$. These are surfaces normal to $\mathbf{k}$. Consider one such surface. The projection of $\mathbf{r}$ onto $\mathbf{k}$

$$r_{\parallel} = \frac{\mathbf{k} \cdot \mathbf{r}}{k}\tag{10.15}$$

from any point on this surface is constant. This is the normal displacement between the origin and the surface. See Fig. 10.1. Rewriting (10.14) in the form

$$\psi_{\mathbf{k}}(\mathbf{r}, t) = A e^{ik[r_{\parallel} - (\omega/k)t]}\tag{10.16}$$

reveals that the rate of increase of $r_{\parallel}$ with respect to a surface of constant $\psi_{\mathbf{k}}$ is the *wave speed*

$$v = \frac{\omega}{k}\tag{10.17}$$

The normalization constant A may be chosen so that

$$\langle \psi_{\mathbf{k}} | \psi_{\mathbf{k}'} \rangle = \langle \varphi_{\mathbf{k}} | \varphi_{\mathbf{k}'} \rangle = \langle \mathbf{k} | \mathbf{k}' \rangle = \iiint \varphi_{\mathbf{k}}^* \varphi_{\mathbf{k}'} dx dy dz = \delta(\mathbf{k} - \mathbf{k}')\tag{10.18}$$

The three-dimensional delta function is defined as the product

$$\delta(\mathbf{r} - \mathbf{r}') = \delta(x - x')\delta(y - y')\delta(z - z')\tag{10.19}$$

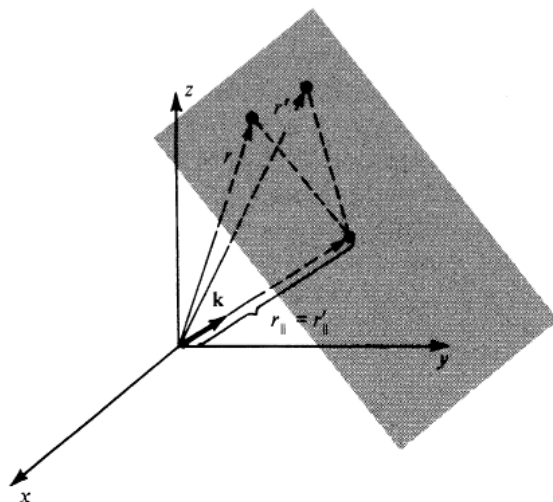


FIGURE 10.1 At any instant of time, the plane wave

$$\psi_{\mathbf{k}}(\mathbf{r}, t) = A \exp[i(\mathbf{k} \cdot \mathbf{r} - \omega t)]$$

is constant on the surface $\mathbf{k} \cdot \mathbf{r} = \text{constant}$. These are surfaces normal to $\mathbf{k}$. At every point $\mathbf{r}$, on such a surface, the projection $r_{\parallel} = \mathbf{k} \cdot \mathbf{r} / k$ is constant.

and has the representation

$$\delta(\mathbf{r} - \mathbf{r}') = \frac{1}{(2\pi)^3} \iiint e^{i\mathbf{k} \cdot (\mathbf{r} - \mathbf{r}')} d\mathbf{k} \quad (10.20)$$

$$d\mathbf{k} = dk_x dk_y dk_z$$

Comparison of this representation with (10.18) yields the normalized wavefunction

$$\varphi_{\mathbf{k}} = \frac{1}{(2\pi)^{3/2}} e^{i\mathbf{k} \cdot \mathbf{r}} \quad (10.21)$$

Superposition of Free-Particle States

A free-particle wave packet may be represented by the superposition

$$\psi(\mathbf{r}, t) = \frac{1}{(2\pi)^{3/2}} \iiint b(\mathbf{k}, t) e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} d\mathbf{k} \quad (10.22)$$

with corresponding inverse

$$b(\mathbf{k}, t) = \frac{1}{(2\pi)^{3/2}} \iiint \psi(\mathbf{r}, t) e^{-i(\mathbf{k} \cdot \mathbf{r} - \omega t)} d\mathbf{r} \quad (10.23)$$

$$d\mathbf{r} = dx dy dz$$

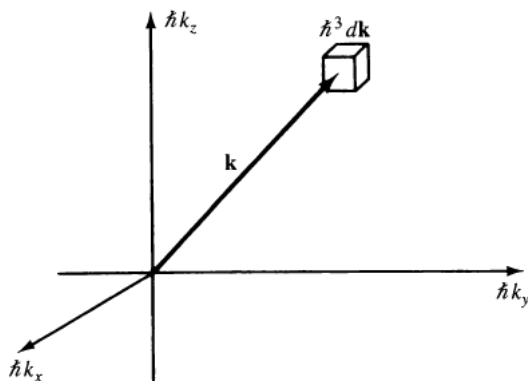


FIGURE 10.2 In the plane-wave decomposition of the free wave packet $\psi(\mathbf{r}, t)$, as given by (10.22), the Fourier amplitude $b(\mathbf{k}, t)$ is such that

$$|b(\mathbf{k}, t)|^2 d\mathbf{k}$$

is the probability that measurement finds the particle with momentum in the volume element $\hbar^3 d\mathbf{k}$ about the value $\hbar\mathbf{k}$.

As in the one-dimensional case, the coefficient b gives the probability

$$P(\mathbf{k}) d\mathbf{k} = |b(\mathbf{k}, t)|^2 d\mathbf{k} \quad (10.24)$$

that measurement at the instant t finds the particle with momentum in the volume element $\hbar^3 d\mathbf{k}$ about the value $\hbar\mathbf{k}$ (Fig. 10.2).

If the probability amplitude $b(\mathbf{k}, t)$ is peaked about a value of $\mathbf{k}$, say $\mathbf{k}_0$, then the three-dimensional wave packet (10.22) propagates with the *group velocity*

$$\mathbf{v}_g = \nabla_{\mathbf{k}} \omega(k) \big|_{\mathbf{k}=\mathbf{k}_0} \quad (10.25)$$

where $\nabla_{\mathbf{k}}$ is written for the gradient with respect to $\mathbf{k}$. Inasmuch as (10.22) depicts the state of a free particle, for each $\mathbf{k}$ -wave component one has

$$E_k = \hbar\omega = \frac{\hbar^2 k^2}{2m} \quad (10.26)$$

This gives $\omega(k)$, which with (10.25) yields

$$\mathbf{v}_g = \frac{\hbar\mathbf{k}_0}{m} = \mathbf{v}_{CL} \quad (10.27)$$

This is the classical velocity of a particle of mass m , moving with momentum $\hbar\mathbf{k}_0$.

For a free particle

$$[\hat{\mathbf{p}}, \hat{H}] = \frac{1}{2m} [\hat{\mathbf{p}}, \hat{p}^2] = 0 \quad (10.28)$$

so that $\hat{\mathbf{p}}$ and $\hat{H}$ have simultaneous eigenstates. These are the functions $\varphi_{\mathbf{k}}(\mathbf{r})$. In the eigenstate $\varphi_{\mathbf{k}}$, the linear momentum $\hbar\mathbf{k}$ and energy $\hbar^2 k^2/2m$ are specified. The state cannot be further resolved. For instance, suppose that we measure the z component of the angular momentum of the particle L_z . This measurement destroys the information in the state before measurement, relating to the linear momentum $\mathbf{p}$. The components of $\mathbf{p}$ and $\mathbf{L}$, in general, do not commute.

What, then, are the states for the free particle, which include specification of L^2 and L_z ? To find these states it proves most convenient to express $\hat{H}$ in spherical coordinates. This is discussed in the next section.

PROBLEMS

- 10.1** If $\psi(\mathbf{r}, t)$ is a free-particle state and $b(\mathbf{k}, t)$ the momentum probability amplitude for this same state, show that

$$\iiint \psi^* \psi d\mathbf{r} = \iiint b^* b d\mathbf{k}$$

- 10.2** At time $t = 0$, a free particle is in the superposition state

$$\psi(\mathbf{r}, 0) = \frac{\pi^{-3/2}}{2} \sin 3x \exp[i(5y + z)]$$

- If the energy of the particle is measured at $t = 0$, what value is found?
 - What possible values of momentum (p_x , p_y , p_z) will measurement find at $t = 0$, and with what probability will these values occur?
 - Given the above state $\psi(\mathbf{r}, 0)$, what is $\psi(\mathbf{r}, t)$?
 - If $\mathbf{p}$ is measured at $t = 0$ and the value $\mathbf{p} = \hbar(3\mathbf{e}_x + 5\mathbf{e}_y + \mathbf{e}_z)$ is found, what is $\psi(\mathbf{r}, t)$?
- 10.3** (a) What is the Hamiltonian for N free, noninteracting particles of mass m ?
 (b) What is the eigenstate of this Hamiltonian corresponding to the eigenvalue

$$E = \frac{\hbar^2}{2m} \sum_{j=1}^N k_j^2$$

- (c) Show that the eigenstate found in part (b) is also an eigenstate of the momentum of the center of mass. What is the velocity of the center of mass in this state?

Answers (partial)

- $\hat{H} = -\sum_{j=1}^N \frac{\hbar^2}{2m} \nabla_j^2$, $\nabla_j = \left(\frac{\partial}{\partial x_j}, \frac{\partial}{\partial y_j}, \frac{\partial}{\partial z_j} \right)$
- $\psi_{\mathbf{k}_1, \mathbf{k}_2, \dots, \mathbf{k}_N} = A \exp \left(i \sum_{j=1}^N \mathbf{k}_j \cdot \mathbf{r}_j \right)$
- $\mathbf{v}_{\text{CM}} = \sum \hbar \mathbf{k}_j / M$

10.2 ■ THE FREE PARTICLE IN SPHERICAL COORDINATES

Hamiltonian

We wish to express the free-particle Hamiltonian (10.2) in spherical coordinates, (r, θ, ϕ) (see Fig. 1.6). We have already found the classical expression for H in spherical coordinates in Chapter 1 [see (1.20)]. Let us again construct this classical form. However, in the present instance we wish H to include the angular momentum term

$$L^2 = (\mathbf{r} \times \mathbf{p})^2 = r^2 p^2 - (\mathbf{r} \cdot \mathbf{p})^2 \quad (10.29)$$

The linear momentum is written $\mathbf{p}$. It follows that

$$H = \frac{p^2}{2m} = \frac{p_r^2}{2m} + \frac{L^2}{2mr^2} \quad (10.30)$$

where p_r is written for the radial component of the particle's momentum.

$$p_r = \frac{1}{r}(\mathbf{r} \cdot \mathbf{p}) \quad (10.31)$$

If we wish to carry (10.29) over to quantum mechanics, we must make sure that all terms in $\hat{H}$ are Hermitian. The two operators in (10.30) are $r^{-2}\hat{L}^2$ and $\hat{p}_r^2$. To examine the Hermiticity of the first operator, we note the following.

Rotation and Angular Momentum

In Section 9.3 we found that the effect of the rotation operator $\hat{R}_{\delta\phi}$ when operating on a function $f(\mathbf{r})$ is to change f by rotating $\mathbf{r}$ to $\mathbf{r} + \delta\phi \times \mathbf{r}$. Suppose that a function f is isotropic² in $\mathbf{r}$; that is, f is independent of the direction of $\mathbf{r}$. It depends only on the magnitude of $\mathbf{r}$. Any function of the form $f(r^2)$ is isotropic in $\mathbf{r}$. For example, $f = ar^2 + br^4$, where a and b are constants, is isotropic in $\mathbf{r}$. What is the value of $f(r^2)$ on the surface of a sphere of radius r_0 ? The answer is, the constant $f(r_0^2)$. An isotropic function is constant on the surface of any sphere about the origin. Now suppose that we operate on an isotropic function with $\hat{R}_{\delta\phi}$. This causes f to change by rotating $\mathbf{r}$ to the value $\mathbf{r} + \delta\phi \times \mathbf{r}$. The new vector lies on the same sphere on which $\mathbf{r}$ lies.

$$(\mathbf{r} + \delta\phi \times \mathbf{r})^2 = r^2 + 2\mathbf{r} \cdot \delta\phi \times \mathbf{r} + O(\delta\phi^2) = r^2$$

Terms of $O(\delta\phi^2)$ are neglected while the middle term vanishes because $\mathbf{r}$ is normal to $\delta\phi \times \mathbf{r}$. It follows that the operator $\hat{R}_{\delta\phi}$ has no effect on $f(r^2)$.

$$\hat{R}_{\delta\phi} f(r^2) = f(r^2) \quad (10.32)$$

²One may also say that f is spherically symmetric.

Since $\hat{R}_{\delta\phi} = (1 + i\delta\phi \cdot \mathbf{L}/\hbar)$, we may conclude that

$$\frac{i\delta\phi \cdot \hat{\mathbf{L}}}{\hbar} f(r^2) = 0 \quad (10.33)$$

In that this statement is true for all axes of rotation about the origin, or equivalently, for all directions of the vector $\delta\phi$, it follows that any isotropic function is a *null eigenstate* of the three components of angular momentum as well as of $\hat{L}^2$.

$$\hat{L}_x f(r^2) = \hat{L}_y f(r^2) = \hat{L}_z f(r^2) = \hat{L}^2 f(r^2) = 0 \quad (10.34)$$

As noted previously in Section 9.2, these spherically symmetric states are called *S* states.

If $g(\mathbf{r})$ is any function of $\mathbf{r}$ (for example, $g = x/r$) and $f(r^2)$ is any isotropic function, then owing to the conclusion immediately above,

$$\begin{aligned} \hat{L}^2 f(r^2) g(\mathbf{r}) &= f(r^2) \hat{L}^2 g(\mathbf{r}) \\ (\hat{L}^2 f(r^2) - f(r^2) \hat{L}^2) g(\mathbf{r}) &= [\hat{L}^2, f(r^2)] g(\mathbf{r}) = 0 \end{aligned}$$

Since this latter equality holds for all differentiable functions g , we obtain

$$[\hat{L}^2, f(r^2)] = 0$$

Similarly,

$$[\hat{L}_x, f(r^2)] = [\hat{L}_y, f(r^2)] = [\hat{L}_z, f(r^2)] = 0 \quad (10.35)$$

We are now prepared to investigate the Hermiticity of the term $r^{-2} \hat{L}^2$ in the Hamiltonian (10.30). With $\hat{r}^{-2}$ denoting multiplication by r^{-2} , we write

$$(\hat{r}^{-2} \hat{L}^2)^\dagger = \hat{L}^{2\dagger} \hat{r}^{-2\dagger} = \hat{L}^2 \hat{r}^{-2} = \hat{r}^{-2} \hat{L}^2 \quad (10.36)$$

so that $\hat{r}^{-2} \hat{L}^2$ is Hermitian. In the last equality we used the fact that $\hat{L}^2$ commutes with the isotropic function r^{-2} .

Radial Momentum

Next we consider the operator

$$\hat{p}_r = r^{-1}(\mathbf{r} \cdot \hat{\mathbf{p}}) = r^{-1}(x\hat{p}_x + y\hat{p}_y + z\hat{p}_z) \quad (10.37)$$

Forming the Hermitian adjoint of $\hat{p}_r$ gives

$$\begin{aligned} (\hat{p}_r)^\dagger &= [r^{-1}(\hat{x}\hat{p}_x + \hat{y}\hat{p}_y + \hat{z}\hat{p}_z)]^\dagger \\ &= (\hat{x}\hat{p}_x)^\dagger (\hat{r}^{-1})^\dagger + (\hat{y}\hat{p}_y)^\dagger (\hat{r}^{-1})^\dagger + (\hat{z}\hat{p}_z)^\dagger (\hat{r}^{-1})^\dagger \end{aligned} \quad (10.38)$$

$$\begin{aligned}
&= \hat{p}_x \hat{x} \hat{r}^{-1} + \dots \\
&\neq \hat{r}^{-1} \hat{x} \hat{p}_x + \dots
\end{aligned}$$

The operators $\hat{r}^{-1}$ and $\hat{x}$ cannot be brought through $\hat{p}_x$. Similarly for the other two terms. We conclude that $\hat{p}_r$ is not Hermitian.

The more appropriate operator corresponding to radial momentum is given by the symmetric form (see Problems 10.5 and 10.6)

$$\hat{p}_r = \frac{1}{2}(\hat{\mathbf{p}}_r + \hat{\mathbf{p}}_r^\dagger) \quad (10.39)$$

or, equivalently,

$$\hat{p}_r = \frac{1}{2} \left(\frac{1}{r} \mathbf{r} \cdot \hat{\mathbf{p}} + \hat{\mathbf{p}} \cdot \mathbf{r} \frac{1}{r} \right) \quad (10.40)$$

The component of $\hat{\mathbf{p}}$ in the direction of $\mathbf{r}$ is given by

$$\frac{1}{r} \mathbf{r} \cdot \hat{\mathbf{p}} = -i\hbar \frac{1}{r} \mathbf{r} \cdot \nabla = -i\hbar \frac{\partial}{\partial r} \quad (10.41)$$

while the second term in $\hat{p}_r$ is given by

$$\hat{\mathbf{p}} \cdot \mathbf{r} \frac{1}{r} = -i\hbar \nabla \cdot \mathbf{e}_r \quad (10.42)$$

where $\mathbf{e}_r$ is written for the unit radius vector, $\mathbf{r}/r$. Let $f(\mathbf{r})$ be a differentiable function of the radius vector $\mathbf{r}$. Consider the operation

$$\begin{aligned}
\hat{p}_r f(r) &= \frac{-i\hbar}{2} \left(\frac{\partial}{\partial r} + \nabla \cdot \mathbf{e}_r \right) f \\
&= \frac{-i\hbar}{2} \left(\frac{\partial f}{\partial r} + \mathbf{e}_r \cdot \nabla f + f \nabla \cdot \mathbf{e}_r \right) \\
&= \frac{-i\hbar}{2} \left(\frac{\partial f}{\partial r} + \frac{\partial f}{\partial r} + \frac{2f}{r} \right) = -i\hbar \left(\frac{\partial f}{\partial r} + \frac{f}{r} \right)
\end{aligned} \quad (10.43)$$

Equivalently, we may write

$$\begin{aligned}
\hat{p}_r f &= -i\hbar \frac{1}{r} \frac{\partial}{\partial r} r f \\
\hat{p}_r &= -i\hbar \frac{1}{r} \frac{\partial}{\partial r} r
\end{aligned} \quad (10.44)$$

With the above definition of $\hat{p}_r$, we may write the following for the Hamiltonian operator:

$$\hat{H} = \frac{\hat{p}_r^2}{2m} + \frac{\hat{L}^2}{2mr^2} \quad (10.45)$$

The student may well ask the following question at this point. We know that the Hamiltonian $\hat{H}$ has the correct representation

$$\hat{H} = \frac{\hat{p}^2}{2m} = -\frac{\hbar^2 \nabla^2}{2m} \quad (10.46)$$

How are we assured that $\hat{H}$, as given by (10.45), with the definition of $\hat{p}_r$ obtained by symmetrization of $\hat{p}_r$, is equivalent to this correct form (10.46)? This question is answered by demonstration. The representation of the Laplacian operator ∇^2 , in spherical coordinates, is

$$\nabla^2 = \frac{1}{r} \frac{\partial^2}{\partial r^2} r + \frac{1}{r^2} \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \sin \theta \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right) \quad (10.47)$$

Noting the equality

$$\left(\frac{1}{r} \frac{\partial}{\partial r} r \right)^2 = \frac{1}{r} \frac{\partial}{\partial r} r \left(\frac{1}{r} \frac{\partial}{\partial r} r \right) = \frac{1}{r} \frac{\partial^2}{\partial r^2} r \quad (10.48)$$

and recalling the expression for $\hat{L}^2$, as given by (9.58), permits the equation

$$\hat{H} = -\frac{\hbar^2 \nabla^2}{2m} = \frac{\hat{p}_r^2}{2m} + \frac{\hat{L}^2}{2mr^2} \quad (10.49)$$

This is the correct form of $\hat{H}$, in spherical coordinates. In the next section we will examine its eigenfunctions and eigenvalues.

PROBLEMS

10.4 What is the time-independent wavefunction in spherical coordinates of a free particle of mass m , zero angular momentum, and energy E which satisfies the property $|r\varphi| = 0$ at $r = 0$? (*Hint*: Introduce the function $u \equiv r\varphi$.)

10.5 (a) Show that

$$[\hat{r}, \hat{p}_r] = i\hbar$$

(b) What properties of φ and ψ insure that

$$\langle \varphi | \hat{p}_r, \psi \rangle = \langle \hat{p}_r \varphi | \psi \rangle$$

$$[\text{Note: } \langle \varphi | \psi \rangle = \int d\Omega \int dr r^2 \varphi^* \psi.]$$

10.6 The current vector $\mathbf{J}$ associated with a wavefunction $\psi(\mathbf{r}, t)$ is given by (7.107)

$$\mathbf{J} = \frac{\hbar}{2mi} (\psi^* \nabla \psi - \psi \nabla \psi^*)$$

The wavefunction $\psi(\mathbf{r}, t)$ may be termed *source-free*, if $\nabla \cdot \mathbf{J} = 0$ for all values of $\mathbf{r}$.

- (a) What is the eigenfunction of $\hat{p}_r$, corresponding to the eigenvalue $\hbar k$?
 (b) Calculate $\nabla \cdot \mathbf{J}$ for this eigenfunction of $\hat{p}_r$.

Answers

- (a) Integration of the eigenvalue equation

$$-i\hbar \frac{1}{r} \frac{\partial}{\partial r} r \tilde{\varphi}_k = \hbar k \tilde{\varphi}_k$$

gives

$$\tilde{\varphi} = A \frac{e^{ikr}}{r}$$

The corresponding time-dependent solution is

$$\tilde{\varphi}_k = \frac{A}{r} e^{i(kr - \omega t)}$$

This “outgoing wave” is a solution to the time-dependent Schrödinger equation for a free particle with no angular momentum. It is important in the construction of scattering states, which will be discussed in Chapter 12.

- (b) The current vector corresponding to $\tilde{\varphi}_k$ only has an r component.

$$\mathbb{J}_r = \frac{|A|^2 (\hbar k / m)}{r^2} \equiv \frac{\Gamma(0)}{4\pi r^2}$$

The divergence of this current is

$$\nabla \cdot \mathbf{J} = \frac{1}{r^2} \frac{\partial}{\partial r} r^2 \mathbb{J}_r = 0 \quad (\text{for } r \neq 0)$$

Since $\mathbf{J}$ is radial and a function only of r , we may write

$$\int_V \nabla \cdot \mathbf{J} d\mathbf{r} = \int_{r=R} \mathbf{J} \cdot d\mathbf{S} = \int_{4\pi} \frac{\Gamma(0)}{4\pi r^2} r^2 d\Omega = \Gamma(0)$$

The spherical volume V has radius R and is centered at the origin, while $d\Omega$ is an element of solid angle about this same origin. Given these two properties of $\nabla \cdot \mathbf{J}$, it follows that

$$\nabla \cdot \mathbf{J} = \Gamma(0) \delta(\mathbf{r})$$

The three-dimensional Dirac delta function is $\delta(\mathbf{r})$ [see (10.19)].

[*Note:* Thus we see that the eigenstates of $\hat{p}_r$ have the unreasonable property of implying that a constant flux of particles, $\Gamma(0)$, emanates from the origin. We may infer from this that the operator $\hat{p}_r$, in spite of its symmetric form (10.39) and proper commutation property with $\hat{r}$ (Problem 10.5), is not a good observable equivalent. Nevertheless, for problems involving a central potential, in quantum mechanics the operator $\hat{p}_r$ proves to be a valuable tool. It is interesting to note that inconsistencies that accompany p_r are also found in classical mechanics. If a free point particle crosses the origin, p_r changes sign instantaneously. This

jump in p_r stems from a choice of coordinate frame. It is in no way associated with a force (the particle is free).^{3,4]}

10.7 Show that the kinetic-energy operator

$$\hat{T} = -\frac{\hbar^2}{2m} \nabla^2$$

is Hermitian for functions in $\mathcal{H}$, (the space of square integrable functions—see Section 4.4). [Hint: Use *Green's theorem*

$$\int_V (f \nabla^2 g - g \nabla^2 f) d\mathbf{r} = \int_S (f \nabla g - g \nabla f) \cdot d\mathbf{S}$$

The volume V is enclosed by the surface S .]

10.8 Show that

$$\nabla \cdot \mathbf{J}(\psi) = 0$$

for the superposition state

$$\psi = \psi_1 + \psi_2$$

provided that

$$\nabla \cdot \mathbf{J}(\psi_1) = \nabla \cdot \mathbf{J}(\psi_2) = 0$$

and

$$\text{Im}(\psi_1^* \nabla^2 \psi_2 + \psi_2^* \nabla^2 \psi_1) = 0$$

10.3 ■ THE FREE-PARTICLE RADIAL WAVEFUNCTION

The time-independent Schrödinger equation for a free particle in spherical coordinates appears as

$$\frac{1}{2m} \left(\hat{p}_r^2 + \frac{\hat{L}^2}{r^2} \right) \varphi_{klm} = E_{klm} \varphi_{klm} \quad (10.50)$$

The quantum number k is defined below. The radial kinetic-energy operator $\hat{p}_r^2/2m$ is inferred from (10.48), while the angular momentum operator $\hat{L}^2$ is given by (9.58). Insofar as $\hat{p}_r^2$ is a function only of r , and $\hat{L}^2$ is a function only of the angle variables (θ, ϕ) , one may seek solution to (10.50) by separation of

³For further discussion of this problem, see R. L. Liboff, I. Nebenzahl, and H. A. Fleishmann, *Am. J. Phys.* **41**, 976 (1973).

⁴Related ambiguities of the radial momentum operator in cylindrical coordinates are discussed in Problems 10.73 and 11.90.

variables. Substituting the product form

$$\varphi_{klm}(r, \theta, \phi) = R_{kl}(r)Y_l^m(\theta, \phi) \quad (10.51)$$

into (10.50) gives

$$\left[-\left(\frac{1}{r} \frac{d^2}{dr^2} r \right) + \frac{l(l+1)}{r^2} \right] R_{kl}(r) = \frac{2mE}{\hbar^2} R_{kl}(r) \quad (10.52)$$

In obtaining (10.52) we have recalled the eigenvalue equation for $\hat{L}^2$ (9.51). With the substitution

$$\begin{aligned} E &\equiv \frac{\hbar^2 k^2}{2m} \\ x &\equiv kr \end{aligned} \quad (10.53)$$

(10.52) becomes the “spherical Bessel differential equation”

$$\frac{d^2}{dx^2} R(x) + \frac{2}{x} \frac{dR(x)}{dx} + \left[1 - \frac{l(l+1)}{x^2} \right] R(x) = 0 \quad (10.54)$$

Spherical Bessel Functions

This ordinary linear equation for the radial function R has two linearly independent solutions.⁵ They are called spherical Bessel and Neumann functions and are denoted conventionally by the symbols $j_l(x)$ and $n_l(x)$, respectively. The first few values of these functions are

$$\begin{aligned} j_0(x) &= \frac{\sin x}{x} & n_0(x) &= -\frac{\cos x}{x} \\ j_1(x) &= \frac{\sin x}{x^2} - \frac{\cos x}{x} & n_1(x) &= -\frac{\cos x}{x^2} - \frac{\sin x}{x} \\ j_2(x) &= \left(\frac{3}{x^3} - \frac{1}{x} \right) \sin x - \frac{3}{x^2} \cos x & n_2(x) &= -\left(\frac{3}{x^3} - \frac{1}{x} \right) \cos x - \frac{3}{x^2} \sin x \end{aligned} \quad (10.55)$$

These functions are sketched in Fig. 10.3, from which it is evident that of the two classes of functions, only the spherical Bessel functions $\{j_l\}$ are regular at the origin. These are the solutions appropriate to the Schrödinger equation (10.50) inasmuch as they are not singular anywhere. Some additional properties of these spherical Bessel and Neumann functions are listed in Table 10.1.

In this manner we find that the eigenstates and eigenenergies of the free-particle Hamiltonian in spherical coordinates are

⁵One obtains these solutions by the method of series substitution. Details may be found in most books on mathematical physics, e.g., G. Goertzel and N. Tralli, *Some Mathematical Methods in Physics*, McGraw-Hill, New York, 1960.

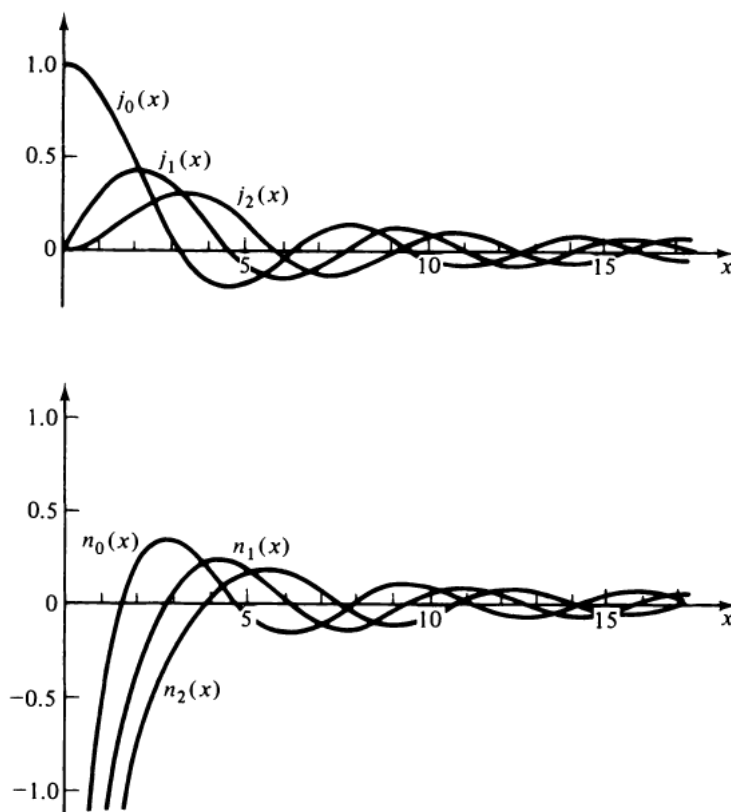


FIGURE 10.3 Spherical Bessel functions $j_l(x)$ and spherical Neumann functions $n_l(x)$ for $l = 0, 1, 2$. Note that only $j_l(x)$ are regular at the origin.

$$\begin{aligned} \varphi_{klm}(r, \theta, \phi) &= j_l(kr) Y_l^m(\theta, \phi) \\ E_k &= \frac{\hbar^2 k^2}{2m} \end{aligned} \quad (10.56)$$

The orthonormality of this sequence $\{\varphi_{klm}\}$ is given by the relation

$$\begin{aligned} \langle lmk | l'm'k' \rangle &= \int_{4\pi} d\Omega [Y_l^m(\theta, \phi)]^* Y_{l'}^{m'}(\theta, \phi) \int_0^\infty j_l(kr) j_{l'}(k'r) r^2 dr \\ &= \delta_{ll'} \delta_{mm'} \frac{\pi}{2k^2} \delta(k - k') \end{aligned} \quad (10.57)$$

The vector $\mathbf{r}$ has the spherical coordinates (r, θ, ϕ) . This orthonormality condition is similar to that corresponding to the free-particle states expressed in Cartesian coordinates (10.18), as well as that corresponding to free-particle motion in one

TABLE 10.1 Properties of the spherical Bessel and Neumann functions*Spherical Bessel Functions*

$$j_l(kr) = \left(-\frac{r}{k}\right)^l \left(\frac{1}{r} \frac{d}{dr}\right)^l j_0(kr)$$

$$j_0(kr) = \frac{\sin kr}{kr}$$

Spherical Neumann Functions

$$n_l(kr) = \left(-\frac{r}{k}\right)^l \left(\frac{1}{r} \frac{d}{dr}\right)^l n_0(kr)$$

$$n_0(kr) = -\frac{\cos kr}{kr}$$

Equation

$$f'' + \frac{2}{x}f + \left[1 - \frac{l(l+1)}{x^2}\right]f = 0$$

Asymptotic Values

$$x \rightarrow 0$$

$$j_l(x) \sim \frac{x^l}{1 \cdot 3 \cdot 5 \cdots (2l+1)}$$

$$x \rightarrow \infty$$

$$j_l(x) \sim \frac{1}{x} \cos \left[x - \frac{\pi}{2}(l+1)\right]$$

$$n_l(x) \sim -\frac{1 \cdot 3 \cdot 5 \cdots (2l-1)}{x^{l+1}}$$

$$n_l(x) \sim \frac{1}{x} \sin \left[x - \frac{\pi}{2}(l+1)\right]$$

Recurrence Relations (f is written for j or n)

$$f_{l-1}(x) + f_{l+1}(x) = (2l+1)x^{-1}f_l(x)$$

$$lf_{l-1}(x) - (l+1)f_{l+1}(x) = (2l+1)\frac{d}{dx}f_l(x)$$

$$\frac{d}{dx}[x^{l+1}j_l(x)] = x^{l+1}j_{l-1}(x)$$

$$\frac{d}{dx}[x^{-l}j_l(x)] = -x^{-l}j_{l+1}(x)$$

Generating Functions

$$\frac{1}{x} \cos \sqrt{x^2 - 2xs} = \sum_0^{\infty} \frac{s^l}{l!} j_{l-1}(x)$$

$$\frac{1}{x} \sin \sqrt{x^2 + 2xs} = \sum_0^{\infty} \frac{(-s)^l}{l!} n_{l-1}(x)$$

Orthogonality

$$\int_0^{\infty} j_l(kr) j_l(k'r) r^2 dr = \frac{\pi}{2k^2} \delta(k - k')$$

$$\int j_1(x) dx = -j_0(x), \quad \int j_0(x) x^2 dx = x^2 j_1(x)$$

$$\int f_l^2(x) x^2 dx = \frac{x^3}{2} [f_l^2(x) - f_{l-1}(x) f_{l+1}(x)]$$

Connection to Bessel and Neumann Functions of Integral Order, J_l and N_l

$$j_l(kr) = \sqrt{\frac{\pi}{2kr}} J_{l+1/2}(kr)$$

$$n_l(kr) = \sqrt{\frac{\pi}{2kr}} N_{l+1/2}(kr)$$

dimension (4.41). In all these cases, the allowed values of momentum, $\hbar k$, comprise a continuum.

Once again we note that the projection

$$\langle r\theta\phi | lmk \rangle = j_l(kr) Y_l^m(\theta, \phi)$$

gives the coordinate representation of the ket vector $|lmk\rangle$. In a similar manner, the coordinate representation of the free-particle ket vector $|\mathbf{k}\rangle$ is given by the projection

$$\langle \mathbf{r} | \mathbf{k} \rangle = \frac{1}{(2\pi)^{3/2}} e^{i\mathbf{k} \cdot \mathbf{r}}$$

(See Problem 7.31.)

Measurements on a Free Particle

Given that a particle is in the eigenstate φ_{klm} , measurement of

$$\begin{aligned} E & \text{ gives } \hbar^2 k^2 / 2m \\ L^2 & \text{ gives } \hbar^2 l(l+1) \\ L_z & \text{ gives } \hbar m \end{aligned} \quad (10.58)$$

How do we know that these values may be measured simultaneously? The answer is that φ_{klm} is a simultaneous eigenfunction of $\hat{H}$, $\hat{L}^2$, and $\hat{L}_z$. The existence of such common eigenfunctions follows from the fact that $\hat{H}$, $\hat{L}^2$, and $\hat{L}_z$ are a commuting set of operators. We have already discussed the commutability of $\hat{L}^2$ and $\hat{L}_z$ in Chapter 9. The fact that these operators commute with $\hat{H}$ follows if they commute with $\hat{p}_r^2$. But $\hat{p}_r^2$ is an isotropic operator; $\hat{p}_r^2 f(r)$ is constant on the surface of any given sphere. It follows that $\hat{p}_r^2$ is unaffected by rotations about the origin; hence

$$[\hat{p}_r^2, \hat{L}_z] = [\hat{p}_r^2, \hat{L}^2] = 0 \quad (10.59)$$

and

$$[\hat{H}, \hat{L}_z] = [\hat{H}, \hat{L}^2] = 0 \quad (10.60)$$

The solution $\varphi_{klm}(r, \theta, \phi)$ should be compared to the eigenstate of the free-particle Hamiltonian in Cartesian coordinates (10.11),

$$\varphi_k(\mathbf{r}) = A e^{i\mathbf{k} \cdot \mathbf{r}} \quad (10.61)$$

Given that a particle is in this state, measurement of

$$\begin{aligned} E & \text{ gives } \frac{\hbar^2 k^2}{2m} \\ p_x & \text{ gives } \hbar k_x \\ p_y & \text{ gives } \hbar k_y \\ p_z & \text{ gives } \hbar k_z \end{aligned} \quad (10.62)$$

In the spherical representation, (L_z, L^2, E) are specified. In the Cartesian representation, $(\mathbf{p}, E)$ are specified. In the latter representation, E is redundant ($E = p^2/2m$), but in the former representation it is not. It is not determined by L^2 and L_z . Thus we find that in either Cartesian or spherical representations, there are three good quantum numbers [recall (1.41)].

Free-Particle S States

The special case $L = 0$ is of interest. For this case the Schrödinger equation (10.50) becomes

$$\left(\frac{\hat{p}_r^2}{2m}\right)\varphi_k = E_k\varphi_k \quad (10.63)$$

The radial kinetic-energy operator $\hat{p}_r^2/2m$ commutes with the radial momentum operator $\hat{p}_r$ and they have common eigenstates. Due to degeneracy, however (the eigenstates E_k are doubly degenerate), eigenfunctions of $\hat{p}_r^2$ are not necessarily eigenfunctions of $\hat{p}_r$ (see Fig. 10.4). Owing to the inadmissibility of the eigenfunctions of $\hat{p}_r$, it is the eigenstates of $\hat{p}_r^2$ alone which are physically relevant. Namely, these functions are

$$\varphi_k = j_0(kr) = \frac{\sin kr}{kr} \quad (10.64)$$

Rewriting φ_k in the form

$$\varphi_k = \tilde{\varphi}_{+k} + \tilde{\varphi}_{-k} = \frac{1}{2i} \left(\frac{e^{ikr}}{kr} - \frac{e^{-ikr}}{kr} \right) \quad (10.65)$$

reveals that it is a superposition of the *outgoing* wave $\tilde{\varphi}_{+k}$, and the *ingoing* wave $\tilde{\varphi}_{-k}$, which gives zero flux at the origin.

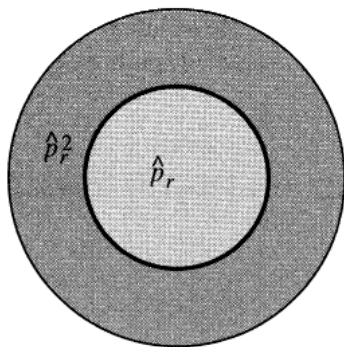


FIGURE 10.4 Central domain represents the eigenstates common to $\hat{p}_r^2$ and $\hat{p}_r$. Peripheral domain represents only eigenstates of $\hat{p}_r^2$, which alone are the physically relevant ones.

Measurement of L_z and L^2 for a Plane Wave

Next, we consider the following important problem. Suppose that a particle of mass m is “prepared” so that it has momentum $\hbar\mathbf{k}$. Then we know that it is in the plane-wave state (10.11).

$$\varphi_{\mathbf{k}} = Ae^{i\mathbf{k}\cdot\mathbf{r}} \quad (10.66)$$

Measurement of E is certain to find $\hbar^2 k^2/2m$. Measurement of momentum is certain to find $\hbar\mathbf{k}$. What will measurement of L_z or L^2 find? And in what states do such measurements leave the particle? To answer this question we must expand the given plane wave in the simultaneous eigenstates of $\hat{H}$, $\hat{L}^2$, and $\hat{L}_z$, that is, φ_{klm} , as given by (10.56). This expansion appears as⁶

$$e^{i\mathbf{k}\cdot\mathbf{r}} = \sum_{l=0}^{\infty} \sum_{m=-l}^l a_{lm}(\mathbf{k}) \varphi_{klm} \quad (10.67)$$

where the coefficients of expansion, a_{lm} , are

$$a_{lm} = 4\pi i^l [Y_l^m(\alpha, \beta)]^* \quad (10.68)$$

and (k, α, β) are the spherical coordinates of $\mathbf{k}$ (Fig. 10.5).

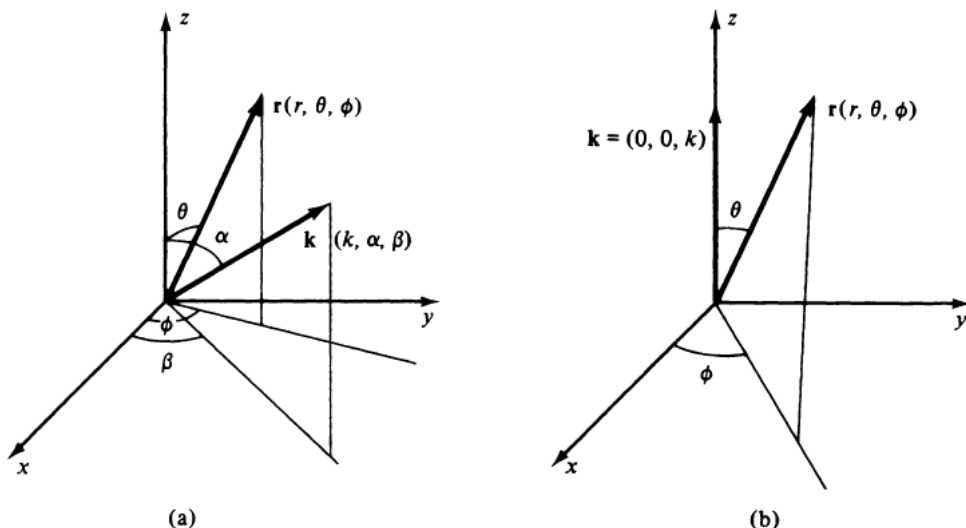


FIGURE 10.5 Coordinates relevant to the expansion of a plane wave in the eigenstates of $\hat{L}^2$ and $\hat{L}_z$. (a) Direction of $\mathbf{k}$ is arbitrary. (b) $\mathbf{k}$ in the direction of the polar axis.

⁶See Goertzel and Tralli, *Some Mathematical Methods in Physics*. This expression is also discussed in Problem 10.11.

The relative probability that measurement of L^2 finds the value $\hbar^2 l(l+1)$ is the partial sum [see (9.82) and (9.83)]

$$p[\hbar^2 l(l+1)] = \sum_{m=-l}^l |a_{lm}|^2 = (4\pi)^2 \sum_{m=-l}^l |Y_l^m(\alpha, \beta)|^2 \quad (10.69)$$

The relative probability that measurement of L^2 finds the value $\hbar m$ is

$$P[\hbar m] = (4\pi)^2 \sum_{l=|m|}^{\infty} |Y_l^m(\alpha, \beta)|^2 \quad (10.70)$$

The Spherical Well

We consider a particle of mass M confined to the interior of a spherical well with impenetrable walls. In the domain $r \geq a$, the wavefunction vanishes. In the domain $r < a$, the time-independent Schrödinger equation is given by (10.50) with general solutions given by (10.56). To impose the boundary condition $\varphi(r=a) = 0$ we set

$$j_l(ka) = 0 \quad (10.71a)$$

With Fig. 10.3 we see that (10.71a) has an infinite number of solutions. To delineate these values we return to the notation $x \equiv kr$ in terms of which (10.71a) becomes

$$j_l(x_{ln}) = 0 \quad (10.71b)$$

where x_{ln} is the n th zero of $j_l(x)$.

Eigenfunctions and eigenenergies for the spherical well are then given by

$$\varphi_{nlm}(r, \theta, \phi) = j_l\left(\frac{x_{ln}r}{a}\right) Y_l^m(\theta, \phi) \quad (10.72a)$$

$$E_{nl} = \frac{\hbar^2 x_{ln}^2}{2Ma^2} = \epsilon \mathbb{R} \left(\frac{a_0}{a}\right)^2 x_{ln}^2 \quad (10.72b)$$

where $\epsilon \equiv m_e/M$. Orthogonality of spherical Bessel functions is given by

$$\int_0^a dr r^2 j_l\left(\frac{x_{ln}r}{a}\right) j_l\left(\frac{x_{ln'}r}{a}\right) = \frac{a^3}{2} [j_{l+1}(x_{ln})]^2 \delta_{nn'} \quad (10.72c)$$

which indicates the nature of normalization of these functions. Note that the continuous spectrum of k values for the free particle in spherical coordinates translates to the discrete spectrum of the quantum number n for the finite spherical well problem.

The following is a table of the first number of zeros of $j_l(x)$.

(l, n)	(0, 1)	(1, 1)	(2, 1)	(0, 2)	(3, 1)	(1, 2)	(4, 1)
x_{ln}	π	4.49	5.76	2π	6.99	7.73	8.18

The ground-state wavefunction and eigenenergy for the spherical well are given by

$$\varphi_G(r, \theta, \phi) = \sqrt{\frac{1}{2\pi a^3}} \frac{1}{j_1(\pi)} j_0\left(\frac{\pi r}{a}\right) \quad (10.73a)$$

$$E_G = \hbar^2 \pi^2 / 2Ma^2 \quad (10.73b)$$

Note that this ground state is nondegenerate.

The Cylindrical Well

We next consider the case of a particle of mass M confined to a cylindrical box of radius a and length L . Expressed in cylindrical coordinates (Fig. 1.7), the potential of this configuration is given by (with r written for ρ as depicted in Fig. 1.7)

$$V(r, z, \phi) = 0, \quad r < a, \quad 0 < z < L \quad (10.74a)$$

$$= \infty \quad \text{elsewhere} \quad (10.74b)$$

Employing the Hamiltonian given in Table 10.2 and the expression for the Laplacian in cylindrical coordinates given in Appendix D, the Schrödinger equation for the confined particle is given by

$$\frac{\partial^2 \varphi}{\partial r^2} + \frac{1}{r} \frac{\partial \varphi}{\partial r} + \frac{1}{r^2} \frac{\partial^2 \varphi}{\partial \phi^2} + \frac{\partial^2 \varphi}{\partial z^2} + k^2 \varphi = 0 \quad (10.75a)$$

$$\hbar^2 k^2 / 2M = E \quad (10.75b)$$

With the separation of coordinates

$$\varphi(r, z, \phi) = R(r)Z(z)\Phi(\phi) \quad (10.76a)$$

(10.75) becomes

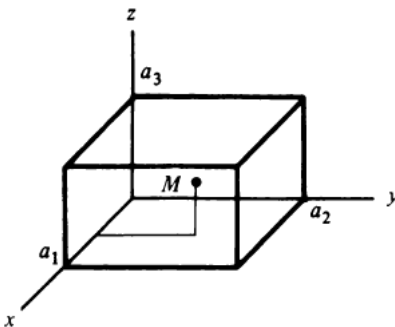
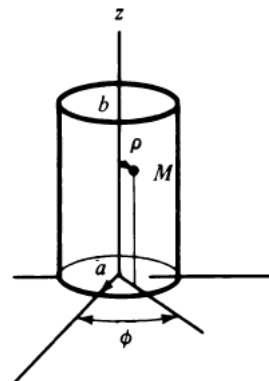
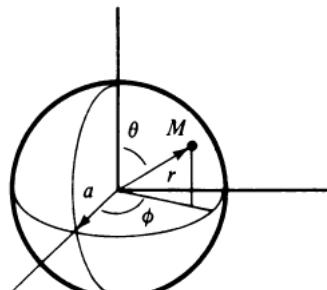
$$\frac{1}{R} \left(\frac{\partial^2 R}{\partial r^2} + \frac{1}{r} \frac{\partial R}{\partial r} \right) + \frac{1}{r^2 \Phi(\phi)} \frac{\partial^2 \Phi}{\partial \phi^2} + \frac{1}{Z} \frac{\partial^2 Z}{\partial z^2} + k^2 = 0 \quad (10.76b)$$

It follows that

$$\frac{1}{Z} \frac{d^2 Z}{dz^2} = \text{constant} \equiv -k_z^2 \quad (10.77a)$$

$$\frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} = \text{constant} \equiv -m^2 \quad (10.77b)$$

TABLE 10.2 Solutions to the three fundamental box problems in quantum mechanics

The Rectangular Box	The Cylindrical Box	The Spherical Box
Edge Lengths a_1, a_2, a_3	Radius a , Height b	Radius a
		
Hamiltonian		
$\hat{H} = (\hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2)/2M$	$\hat{H} = (\hat{p}_\rho^2 + \hat{p}_z^2 + \hat{L}_z^2/\rho^2)/2M$	$\hat{H} = (\hat{p}_r^2 + \hat{L}^2/r^2)/2M$
$\hat{p}_x^2 = \left(-i\hbar \frac{\partial}{\partial x}\right)^2$	$\hat{p}_\rho^2 = -\hbar^2 \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial}{\partial \rho}\right)$	$\hat{p}_r^2 = -\hbar^2 \left(\frac{1}{r} \frac{\partial}{\partial r} r\right)^2$
Eigenfunction		
$\varphi_{qst} = A_{qst} \sin k_q x \sin k_s y \sin k_t z$	$\varphi_{qmn} = A_{qmn} J_m(K_{mn}\rho) \sin k_q z e^{im\phi}$	$\varphi_{nlm} = A_{nlm} j_l(k_{ln}r) Y_l^m(\theta, \phi)$
$(A_{qst})^2 = 8/a_1 a_2 a_3$	$(A_{qmn})^2 = 2/\pi b [a J_m'(K_{mn}a)]^2$	$(A_{nlm})^2 = 2/a^3 [j_l'(k_{ln}a)]^2$
$\sin k_q a_1 = \sin k_s a_2 = \sin k_t a_3 = 0$	$\sin k_q b = J_m(K_{mn}a) = 0$	$j_l(k_{ln}a) = 0$
Wave Equation	Bessel's Equation	Spherical Bessel Equation
$\left(\frac{d^2}{dx^2} + k^2\right) \sin kx = 0$	$\left[\frac{1}{x^2} \left(x \frac{d}{dx}\right)^2 + 1 - \frac{m^2}{x^2}\right] J_m(x) = 0$	$\left[\left(\frac{1}{x} \frac{d}{dx} x\right)^2 + 1 - \frac{l(l+1)}{x^2}\right] j_l(x) = 0$
Eigenenergy		
$E_{qst} = \hbar^2(k_q^2 + k_s^2 + k_t^2)/2M$	$E_{qmn} = \hbar^2(K_{mn}^2 + k_q^2)/2M$	$E_{nl} = \hbar^2 k_{ln}^2/2M, ak_{ln} \equiv x_{ln}$

$$\frac{1}{R}(r^2 R'' + r R') + r^2(k^2 - k_z^2) = m^2 \quad (10.77c)$$

where a prime denotes differentiation with respect to r . With conditions (10.74) we find

$$Z(z) = A \sin k_z z, \quad k_z L = n_z \pi, \quad n_z = 1, 2, \dots \quad (10.78a)$$

Furthermore, as $\Phi(\phi) = \Phi(\phi + 2\pi)$, we obtain

$$\Phi(\phi) = B e^{im\phi}, \quad m = 0, \pm 1, \pm 2, \dots \quad (10.78b)$$

where A and B are constants. Returning to (10.77c) and labeling

$$\begin{aligned} k^2 - k_z^2 &\equiv K^2 \\ \rho &\equiv Kr \end{aligned} \quad (10.78c)$$

there results

$$\rho^2 R'' + \rho R' + (\rho^2 - m^2)R = 0 \quad (10.79)$$

which is known as *Bessel's equation*. General solutions to this equation are given by

$$R(\rho) = C_1 J_m(\rho) + C_2 N_m(\rho) \quad (10.80)$$

where C_1 and C_2 are constants. The functions $J_m(\rho)$ and $N_m(\rho)$ are called *Bessel* and *Neumann functions of the first kind*, respectively. As $N_m(0) = -\infty$, it is omitted in (10.80). The functions $J_m(\rho)$ oscillate.⁷ Graphs of these functions are similar to those for the spherical Bessel functions shown in Fig. 10.3. The remaining boundary conditions [first of (10.74a)] gives

$$R(r = a) = 0 = C_1 J_m(aK) \quad (10.81a)$$

Let us call the s th finite zero of $J_m(\rho)$, x_{ms} so that

$$J_m(aK_{ms}) \equiv J_m(x_{ms}) = 0 \quad (10.81b)$$

The following is a table of the first few zeros of $J_n(\rho)$.

	$s = 1$	$s = 2$	$s = 3$
x_{0s}	2.40	5.52	8.65
x_{1s}	3.83	7.02	10.17
x_{2s}	5.14	8.42	11.62
x_{3s}	6.30	9.76	13.02

Eigenenergies are given by

$$E = \frac{\hbar^2 k^2}{2M} = \frac{\hbar^2}{2M} (K^2 + k_z^2) = \frac{\hbar^2}{2M} \left[K^2 + \left(\frac{m\pi}{L} \right)^2 \right]$$

⁷For integer m , $J_{-m}(\rho) = (-1)^m J_m(\rho)$, so that with no loss in generality, we consider only $m \geq 0$.

With $x_{ms} \equiv aK_{ms}$, the preceding becomes

$$E_{msn_z} = \frac{\hbar^2}{2M} \left[\frac{x_{ms}^2}{a^2} + \left(\frac{n_z \pi}{L} \right)^2 \right] \quad (10.82a)$$

Corresponding eigenfunctions are given by

$$\begin{aligned} \varphi_{msn_z}(r, z, \phi) &= A J_m \left(\frac{r x_{ms}}{a} \right) \sin \left(\frac{n_z \pi z}{L} \right) e^{im\phi} \\ m &\geq 0, \quad s > 0, \quad n_z \geq 1 \end{aligned} \quad (10.82b)$$

All three parameters are integers.

The ground state of this system is given by

$$E_G = \frac{\hbar^2}{2M} \left[\frac{(2.40)^2}{a^2} + \frac{\pi^2}{L^2} \right] \quad (10.83a)$$

$$\varphi_G(r, z, \phi) = A J_0 \left(\frac{2.40r}{a} \right) \sin \left(\frac{\pi z}{L} \right) \quad (10.83b)$$

and is positive within the cylindrical box. Orthogonality of the Bessel functions in (10.82b) parallels that of (10.72c) and is given by

$$\int_0^a dr r J_m \left(\frac{r x_{ms}}{a} \right) J_m \left(\frac{r x_{ms'}}{a} \right) = \frac{a^2}{2} [J_{m+1}(x_{ms})]^2 \delta_{ss'} \quad (10.84)$$

which, again, indicates the nature of normalization of these functions. A compilation of properties of the rectangular, spherical, and cylindrical quantum wells is listed in Table 10.2.

Note that for spherical Bessel functions (10.72a), x_{ln} represents the n th zero of $j_l(x)$ and l is orbital quantum number. For Bessel functions of the first kind (10.82b), x_{ms} represents the s th zero of $J_m(x)$ and m is azimuthal quantum number [see (9.51)]. Thus, for both spherical and cylindrical quantum wells, the orders of respective Bessel functions that enter the analysis are angular-momentum quantum numbers. Note that for $\hat{H}_{\text{cyl}}(r, z, \phi)$, $[\hat{L}_z, \hat{H}_{\text{cyl}}] = 0$, whereas for $H_{\text{sph}}(r, \theta, \phi)$, $[\hat{L}^2, \hat{H}_{\text{sph}}] = [\hat{L}_z, \hat{H}_{\text{sph}}] = 0$. For the latter case, l and m are good quantum numbers, corresponding to invariance of the Hamiltonian to rotations that leave r^2 invariant. For the cylindrical case, only m is a good (angular momentum) quantum number, corresponding to invariance of the Hamiltonian to rotations about the z axis.

PROBLEMS

- 10.9** Calculate the divergence of particle current, $\nabla \cdot \mathbf{J}$, for a collection of particles that are all in the state

$$\psi(r, t) = j_2(kr)e^{-i\omega t}$$

$$\hbar\omega = \frac{\hbar^2 k^2}{2m}$$

- 10.10** A spherically propagating shell contains N neutrons, which are all in the state

$$\psi(\mathbf{r}, 0) = 4\pi i \left[\frac{\sin kr}{(kr)^2} - \frac{\cos kr}{kr} \right] \frac{3Y_1^0(\theta, \phi) + 5Y_1^{-1}(\theta, \phi)}{\sqrt{34}}$$

at $t = 0$.

- (a) What is $\psi(\mathbf{r}, t)$?
- (b) What is the expectation of the energy for this “beam”?
- (c) What possible values of L^2 and L_z will measurement find and how many neutrons will have these values?
- (d) If at $t = 0$, measurement of L^2 finds the value $2\hbar^2$, what is $\psi(\mathbf{r}, t)$?
- (e) If at $t = 0$, measurement of L_z finds the value $-\hbar$, what is $\psi(\mathbf{r}, t)$?

- 10.11** Use the expansion of a plane wave in spherical harmonics,

$$e^{i\mathbf{k} \cdot \mathbf{r}} = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l i^l j_l(kr) [Y_l^m(\alpha, \beta)]^* Y_l^m(\theta, \phi)$$

together with the spherical coordinate representation of $\delta(\mathbf{r} - \mathbf{r}')$,

$$\delta(\mathbf{r} - \mathbf{r}') = \frac{\delta(r - r') \delta(\theta - \theta') \delta(\phi - \phi')}{r^2 \sin \theta}$$

$$= \left(\frac{1}{2\pi} \right)^3 \int_0^{2\pi} \int_0^\pi \int_0^\infty e^{i\mathbf{k} \cdot (\mathbf{r} - \mathbf{r}') k^2} dk \sin \alpha d\alpha d\beta$$

(the spherical coordinates of $\mathbf{k}$ are k, α, β ; see Fig. 10.5a) to obtain the orthonormality condition

$$\delta(\mathbf{r} - \mathbf{r}') = \frac{2}{\pi} \sum_l \sum_m [Y_l^m(\theta', \phi')]^* Y_l^m(\theta, \phi) \int_0^\infty j_l(kr) j_l(kr') k^2 dk$$

The spherical coordinates of $\mathbf{r}$ are (r, θ, ϕ) , and those of $\mathbf{r}'$ are (r', θ', ϕ') . [Compare with (C.14) in Appendix C.]

- 10.12** Use the addition theorem for spherical harmonics (see Fig. 9.16) to reduce the first equation of Problem 10.11 to the expansion

$$e^{ikz} = e^{ikr \cos \theta} = \sum_{l=0}^{\infty} (2l+1) i^l j_l(kr) P_l(\cos \theta)$$

Note that in this description, $\mathbf{k}$ is aligned with the polar (z) axis, so that $\mathbf{k} \cdot \mathbf{r} = kz$ (see Fig. 10.5b). This expansion is important to the theory of partial wave scattering and will be called upon in Chapter 14.

- 10.13** The expansion in Problem 10.12 of the plane wave e^{ikz} indicates that the probability of measuring $L^2 = \hbar^2 l(l+1)$ is

$$P[\hbar^2 l(l+1)] \simeq (2l+1)^2$$

Give a semiclassical heuristic argument in support of this conclusion (i.e., that $P \sim l^2$).

Answer

Consider a surface S , of constant phase of the plane wave, $\exp(ikz)$. (See Fig. 10.6.) All points in the annular region $dS = 2\pi r_\perp dr_\perp = \pi d(r_\perp^2)$ correspond to angular momentum $L = r_\perp p_z = r_\perp \hbar k$. It follows that

$$dS = \pi d\left(\frac{L^2}{\hbar^2 k^2}\right)$$

The probability of finding such “points” is proportional to the annular surface dS , so (k^2 is constant)

$$dP \sim dS \sim dL^2$$

In the classical (correspondence) limit, $L^2 \sim \hbar^2 l^2$ and $P \sim l^2$.

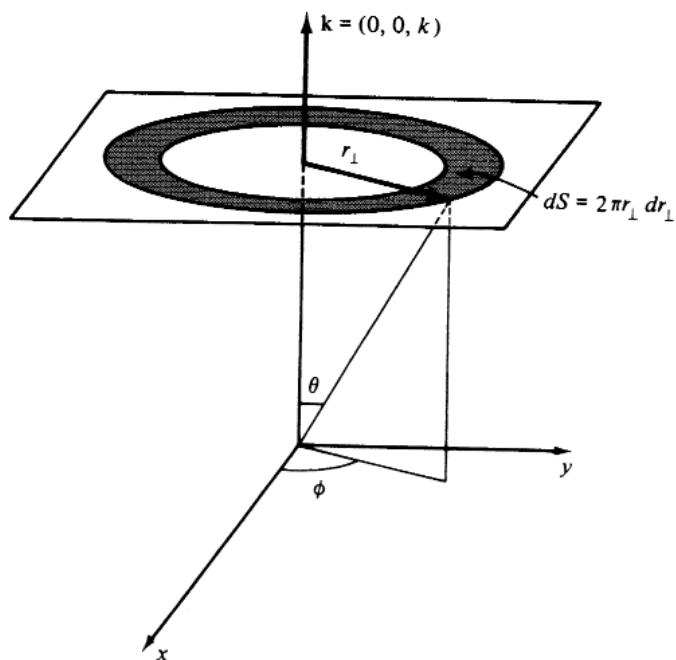


FIGURE 10.6 The probability of finding a particle in a plane-wave state with angular momentum $\sim \hbar l$ increases as l^2 . (See Problem 10.13.)

- 10.14** How many independent eigenstates are there corresponding to a free particle moving with energy $E_k = \hbar^2 k^2 / 2m$ in

- (a) the Cartesian coordinate representation?
 (b) the spherical coordinate representation?

Give a classical description of the different orbits corresponding to these degenerate states.

Answers (partial)

- (a) In the Cartesian representation, any state

$$\varphi_{\mathbf{k}} = A e^{i\mathbf{k} \cdot \mathbf{r}}, \quad k^2 = \frac{2mE}{\hbar^2}$$

is an eigenstate corresponding to the given value of E . These $\mathbf{k}$ vectors describe a sphere of radius $\sqrt{2mE/\hbar^2}$. This continuum of states corresponds to aiming the particle in different directions, while holding its speed, $\hbar k/m$, fixed.

- (b) In the spherical representation, any state

$$\varphi_{klm} = j_l(kr) Y_l^m(\theta, \phi), \quad k^2 = \frac{2mE}{\hbar^2}$$

is an eigenstate corresponding to the given value of E . Different states are obtained by choosing different values of l and m . This countable infinity of states corresponds to propitious choice of straight-line trajectories about the origin, all at constant speed, $\hbar k/m$.

- 10.15** An electron is confined to a spherical well of radius $a = 10 \text{ \AA}$. Calculate the emission frequency, f_1 (in Hz), of radiation for this system due to transition from the first excited state to the ground state. State the class of radiation to which this frequency belongs (IR, visible, UV, soft x rays, etc.).
- 10.16** A “quantum dot,” relevant to semiconductor devices, may be modeled as an electron in a spherical well with perfectly reflecting walls. Design a quantum dot whose characteristic frequency of emission is 10 GHz, where “characteristic frequency” corresponds to decay from the first excited state to the ground state. That is, obtain the radius, a , of the spherical cavity that has this property. Repeat this problem for the quantum dot that is a cube of edge length $2b$. In either case assume an effective mass of $m = 0.067m_e$. Give your answers in μm .
- 10.17** (a) What are the eigenfunctions and eigenenergies (in eV) of the ground and first excited states of an electron trapped in a cylindrical box of radius $a = 1.5 \text{ \AA}$ and length $L = 3.0 \text{ \AA}$? What is the degeneracy of these two eigenenergies?
- (b) An electron propagates in a “quantum wire” of circular cross-sectional radius $5.2 \text{ \AA}$ and z component of kinetic energy 1.03 eV. What is the lowest energy wavefunction and energy (in eV) corresponding to this condition?
- (Note: A quantum wire is a rectilinear wirelike domain of narrow band-gap material immersed in a medium of wide band-gap material. In this problem we model the quantum wire as being of infinite length and assume that the propagating electron acts as a confined free particle.)

10.4 ■ A CHARGED PARTICLE IN A MAGNETIC FIELD

A closely allied motion to that of a free particle is the motion of a charged particle (e.g., an electron) in a uniform, constant magnetic field $\mathbf{B}$. The Hamiltonian for the electron is given by

$$H = \frac{1}{2m} \left(\mathbf{p} - \frac{e}{c} \mathbf{A} \right)^2 \quad (10.85)$$

The magnetic field is related to the vector potential $\mathbf{A}$ through the relation

$$\mathbf{B} = \nabla \times \mathbf{A}$$

The Cartesian components of $\mathbf{A}$,

$$\mathbf{A} = (-yB, 0, 0)$$

generate a uniform magnetic field which points in the z direction.

$$\mathbf{B} = (0, 0, B)$$

Substituting this value of $\mathbf{A}$ into the Hamiltonian above gives the time-independent Schrödinger equation

$$\hat{H}\varphi = \left[\frac{1}{2m} \left(\hat{p}_x + \frac{eyB}{c} \right)^2 + \frac{\hat{p}_y^2}{2m} + \frac{\hat{p}_z^2}{2m} \right] \varphi = E \quad (10.86)$$

Since the coordinates x and z are missing from the Hamiltonian, it follows that

$$[\hat{p}_x, \hat{H}] = [\hat{p}_z, \hat{H}] = 0$$

and we may conclude that $\hat{p}_x$, $\hat{p}_z$, and $\hat{H}$ have simultaneous eigenstates. The eigenstates of $\hat{p}_x$ and $\hat{p}_z$ appear as

$$\varphi_{k_x k_z} = e^{i(k_x x + k_z z)}$$

so that we may write the common eigenstates of $\hat{H}$, $\hat{p}_x$, and $\hat{p}_z$ in the form

$$\varphi = e^{i(k_x x + k_z z)} f(y) \quad (10.87)$$

Substituting this product into (10.86) gives

$$\left[\frac{\hat{p}_y^2}{2m} + \frac{K}{2} (y - y_0)^2 \right] f = \left(E - \frac{\hbar^2 k_z^2}{2m} \right) f \quad (10.88)$$

where we have set

$$y_0 \equiv -\frac{c\hbar k_x}{e\mathcal{B}}$$

$$\frac{K}{m} \equiv \left(\frac{e\mathcal{B}}{mc}\right)^2 \equiv \Omega^2$$

The frequency Ω is called the *cyclotron frequency*. This is the frequency of rotation corresponding to the classical motion of a charged particle in a uniform magnetic field (see Problem 10.18).

The Schrödinger equation (10.88) is the same as that for a simple harmonic oscillator constrained to move along the y axis, about the point y_0 , with natural frequency Ω . From Section 7.2 we recall that the eigenenergies of this equation are

$$E_n - \frac{\hbar^2 k_z^2}{2m} = \hbar\Omega \left(n + \frac{1}{2}\right)$$

which gives the desired result

$$E_n = \hbar\Omega \left(n + \frac{1}{2}\right) + \frac{\hbar^2 k_z^2}{2m} \quad (10.89)$$

The kinetic-energy term $\hbar^2 k_z^2 / 2m$ corresponds to free, linear motion parallel to the z axis. Classically, such motion is unaffected by a magnetic field in the z direction. The first term in E_n corresponds to the rotational motion normal to the $\mathcal{B}$ field. In the corresponding classical motion the charged particle moves in a helix of constant radius, constant energy, constant rotational frequency, and constant z velocity. The projection of the motion onto the xy plane is a circle with a fixed center (Fig. 10.7). The energy levels (10.89) are commonly referred to as *Landau levels*.

The eigenfunction corresponding to the eigenenergy (10.89) is

$$f_n = A_n \mathcal{H}_n \left[\sqrt{\frac{m\Omega}{\hbar}} (y - y_0) \right] \exp \left[-\frac{1}{2} \frac{m\Omega}{\hbar} (y - y_0)^2 \right]$$

[recall (7.59)]. The n th-order Hermite polynomial is written $\mathcal{H}_n$, while A_n is a normalization constant. Together with (10.87), this form for f_n gives the wavefunction

$$\varphi_n = A_n \mathcal{H}_n \left[\sqrt{\frac{m\Omega}{\hbar}} (y - y_0) \right] \exp \left[-\frac{1}{2} \frac{m\Omega}{\hbar} (y - y_0)^2 + i(k_x x + k_z z) \right]$$

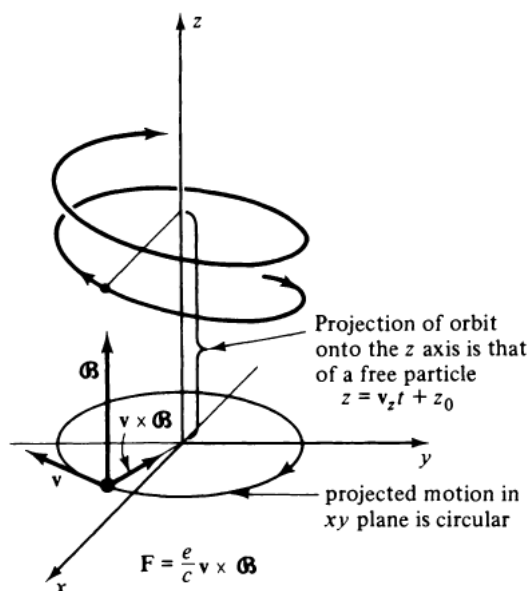


FIGURE 10.7 Helical motion of a positive charge in a uniform, constant magnetic field that points in the z direction.

for a charged particle moving in a uniform magnetic field which points in the z direction.

Degeneracy of Landau Levels

In this section we wish to discover the manner in which the continuous energy spectrum of a free electron changes when the electron is in the presence of a magnetic field. In the course of this discussion we will obtain an expression for the degeneracy of Landau levels.

To examine this problem, we consider the electron to be enclosed in a large cubical box of edge length L . Free-particle wavefunctions and energies are given by (10.14). To account for the finite enclosure, for L sufficiently large, say $L^2 \gg \hbar/m\Omega$, we impose *periodic boundary conditions*:

$$\varphi(x, y, z) = \varphi(x + L, y + L, z + L)$$

As the Landau levels given by (10.89) are evidently degenerate in k_x , we will focus on this wavenumber for the free-particle motion. Substituting the free-particle state (10.14) into the relation above gives

$$k_x = \frac{2\pi n_x}{L}$$

$$n_x = 0, \pm 1, \pm 2, \dots$$

Thus we discover the following important fact. The continuous energy spectrum of a free, unconfined particle changes to a nearly continuous discrete spectrum when the particle is confined in a large enclosure.

In the presence of a magnetic field, the Schrödinger equation for the electron is given by (10.88). As noted in Problem 10.18, y_0 is associated with the center of the corresponding classical circular motion. Thus, for the present case we may take y_0 to lie between 0 and L .

It follows that the maximum value of this parameter $\bar{y}_0$ is given by

$$\bar{y}_0 = L = \frac{c\hbar\bar{k}_x}{|e|\mathcal{B}} = \frac{ch\bar{n}_x}{|e|\mathcal{B}L}$$

We may conclude that n_x values are positive numbers and have the maximum value⁸

$$\bar{n}_x \equiv \bar{g} = \left(\frac{|e|\mathcal{B}}{c} \right) L^2 \quad (10.91)$$

This is the desired expression for the degeneracy $\bar{g}$ of a Landau level. The presence of L^2 in (10.91) corresponds to the property that a given helical orbit can be displaced anywhere in the xy plane without changing the energy of the electron (see Fig. 10.7).

Thus, we find that the energy spectrum of a confined electron changes from a nearly continuous one for $\mathcal{B} = 0$ to a discrete spectrum for $\mathcal{B} > 0$, with degeneracy given by (10.91).

From (10.89) we see that the spacing between Landau levels, at fixed k_z , is the constant value

$$\Delta E = \hbar\Omega$$

The degeneracy $\bar{g}$ given by (10.91) gives the number of free-particle states that contribute to the increment ΔE . See Fig. 10.8. Note, in particular, the resemblance between the equally spaced Landau levels and the equally spaced levels of the harmonic oscillator shown in Fig. 7.8. This congruence of spectra stems from the previously described parallel structure of the two respective Hamiltonians. In either event, the density of states, $g(E)$ (Section 8.8), when plotted as a function of energy, is a series of equally spaced delta functions.

Note further that in the classical limit $\hbar \rightarrow 0$, the degeneracy $\bar{g}$ grows infinite and the spacing between levels ΔE goes to zero.

Fermi Energy and Landau Levels

Application of the preceding results may be made to the conduction electrons in a two-dimensional metal. In this event, the degenerate states shown in Fig. 10.8a become filled with electrons. The highest energy state occupied (at 0 K) is that of

⁸We employ the barred variable $\bar{g}$ to distinguish it from the unbarred variable g used to denote density of states elsewhere in the text.

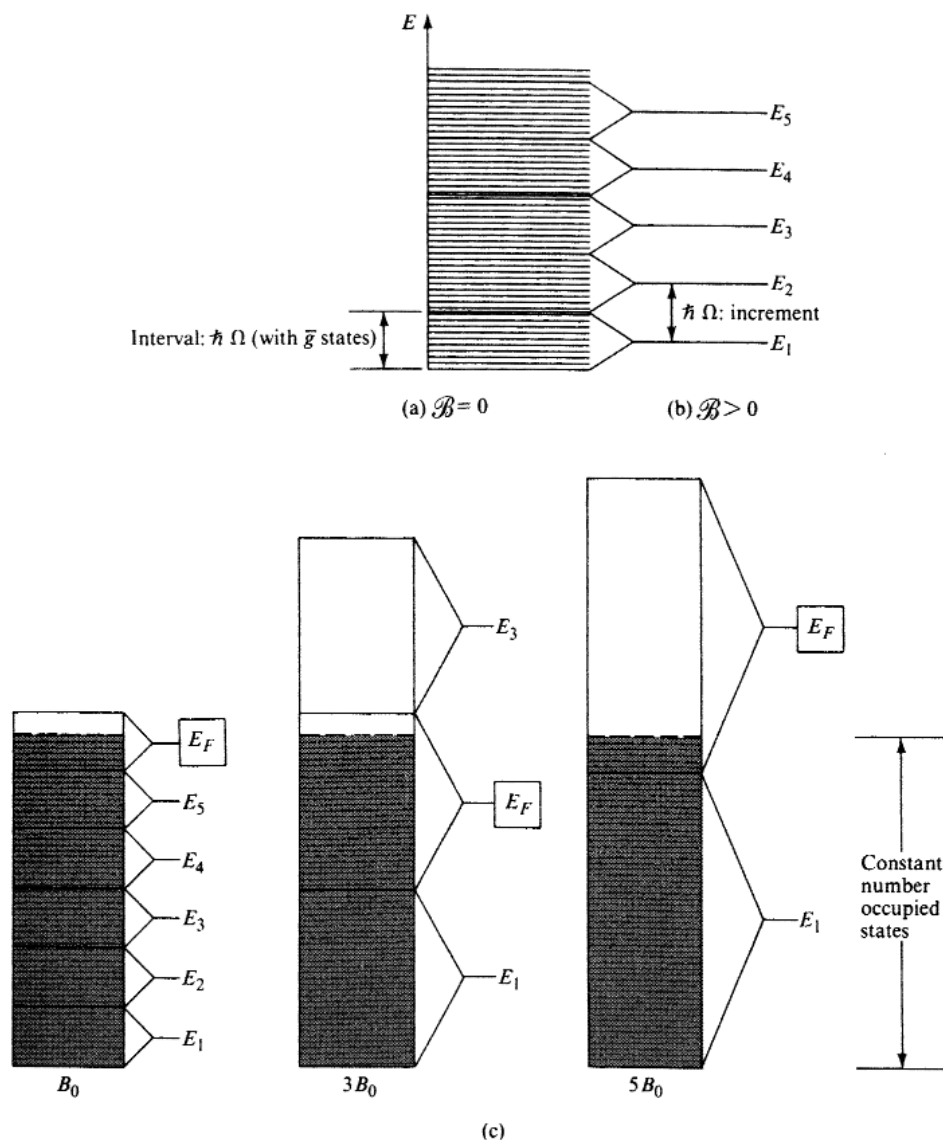


FIGURE 10.8 (a) Nearly continuous discrete spectrum for a particle confined to a large box with $B = 0$. (b) Equally spaced Landau levels corresponding to $B > 0$. Each increment of energy, $\hbar\Omega$, corresponds to $\bar{g}$ free-particle states, which, in turn, is the degeneracy of each Landau level. (c) Variation of the Fermi energy with change in B .

the Fermi energy, E_F [Sections 2.3 (Fig. 2.3), 8.4, 12.9]. With a magnetic field B normal to the plane of the sample, we may identify E_F as the highest Landau level occupied. How does E_F change with change in B ? To answer this question, we first note that the maximum number of electrons at the Fermi level is given by the degeneracy factor $\bar{g}$ (10.91). Let $n^{\text{oc}} \leq \bar{g}$ denote the number of occupied

states in this level. The value of E_F remains constant as long as $n^{oc} > 0$. When $\mathcal{B}$ is increased, E_F changes to the value of the new partially occupied Landau level. See Fig. 10.8c. Such variation of E_F exhibits a periodicity with respect to $\mathcal{B}^{-1}$. This phenomenon is a component of the *de Haas–van Alphen effect*.⁹

PROBLEMS

10.18 The following is a problem in classical physics. The force on a charged particle in a uniform magnetic field $\mathcal{B}$ is

$$\mathbf{F} = \frac{d}{dt}(m\mathbf{v}) = \frac{e}{c}\mathbf{v} \times \mathcal{B}$$

with $\mathcal{B} = (0, 0, \mathcal{B})$.

(a) Show that

$$\frac{1}{2}mv^2 = \text{constant}$$

(b) Show that

$$p_z = mv_z = \text{constant}$$

(c) Show that the motion of the particle is that of a helix whose axis is parallel to $\mathcal{B}$ and whose projection onto the xy plane is circular with constant angular frequency Ω .

(d) Show that the center of this circle in the xy plane has coordinates

$$y_0 = \frac{-cp_x}{e\mathcal{B}} + y = \frac{-cmv_x}{e\mathcal{B}} + y$$

$$x_0 = \frac{cmv_y}{e\mathcal{B}} + x = \frac{cp_y}{e\mathcal{B}} + x$$

Note that p_x , canonical momentum, is not equal to mv_x for $\mathbf{A} = (A_x, 0, 0)$. The correct relation follows from (1.14) and (10.85).

10.19 Show that the operator

$$\hat{x}_0 \equiv \hat{x} + \frac{c\hat{p}_y}{e\mathcal{B}}$$

commutes with $\hat{H}$ as given in (10.86) but does not commute with

$$\hat{y}_0 = \frac{-c\hat{p}_x}{e\mathcal{B}}$$

These operators correspond to the coordinates of the center of the related projected classical motion in the xy plane. In quantum mechanics we see that although x_0 and

⁹For further discussion, see C. Kittel, *Introduction to Solid State Physics*, 6th ed., Wiley, New York, 1986, Chap. 9.

E , or y_0 and E , may, respectively, be specified simultaneously, x_0 and y_0 may not be simultaneously specified.

- 10.20** (a) What is the vector potential $\mathbf{A}$ which gives the uniform $\mathcal{B}$ field $(0, 0, \mathcal{B})$ which includes $A_x = A_z = 0$?
- (b) What is the form of the wavefunctions φ_n corresponding to this choice of vector potential? How do they compare to the wavefunctions corresponding to $A_y = A_z = 0$ found in the text?
- (c) How do the eigenenergies compare to those found in the representation $A_y = A_z = 0$?
- 10.21** What is the nature of the frequency spectrum emitted by a charged particle moving in a uniform magnetic field? (Assume that the kinetic energy parallel to $\mathcal{B}$ does not change.) For an electron moving in a $\mathcal{B}$ field of 10^4 gauss, what type of radiation is this (x rays, microwaves, etc.)?

10.5 ■ THE TWO-PARTICLE PROBLEM

Coordinates Relative to the Center of Mass

When dealing with systems containing more than one particle (e.g., an atom), it is convenient to separate the motion into that of the center of mass of the system and motion relative to the center of mass. This separation is effected through a partitioning of the Hamiltonian into a part H_{CM} , involving center of mass coordinates, and a part, H_{rel} , containing coordinates relative to the center of mass.

For example, consider the two-particle Hamiltonian

$$H = \frac{p_1^2}{2m_1} + \frac{p_2^2}{2m_2} + V(|\mathbf{r}_1 - \mathbf{r}_2|) \quad (10.92)$$

The potential of interaction $V(|\mathbf{r}_1 - \mathbf{r}_2|)$ is a function only of the radial distance between the particles. For instance, for the hydrogen atom, the interaction V is the Coulomb potential

$$V = -\frac{e^2}{r} \quad (10.92a)$$

where we have written r for the distance between particles, $|\mathbf{r}_1 - \mathbf{r}_2|$. Such potentials, which are only a function of the scalar distance r , are called *central potentials*.

In the Hamiltonian above, $\mathbf{p}_1$ and $\mathbf{p}_2$ are the linear momenta of particle 1 and particle 2, respectively, while m_1 and m_2 are the respective masses of these particles.

A two-particle system has six degrees of freedom which are the number of coordinates in the state vector $(\mathbf{r}_1, \mathbf{p}_1; \mathbf{r}_2, \mathbf{p}_2)$. The partitioning of the Hamiltonian into $H_{CM} + H_{rel}$ is generated through the transformation of variables

$$(\mathbf{r}_1, \mathbf{p}_1; \mathbf{r}_2, \mathbf{p}_2) \rightarrow (\mathbf{r}, \mathbf{p}; \mathcal{R}, \mathcal{P}) \quad (10.93)$$

where

$$\begin{aligned}\mathbf{r} &= \mathbf{r}_2 - \mathbf{r}_1, & \mathcal{P} &= \mathbf{p}_1 + \mathbf{p}_2 \\ \mathbf{p} &= \frac{m_1 \mathbf{p}_2 - m_2 \mathbf{p}_1}{m_1 + m_2}, & \mathcal{R} &= \frac{m_1 \mathbf{r}_1 + m_2 \mathbf{r}_2}{m_1 + m_2}\end{aligned}\quad (10.94)$$

Using these equations the Hamiltonian (10.92) is transformed to the sum

$$H = \frac{\mathcal{P}^2}{2M} + \left[\frac{p^2}{2\mu} + V(r) \right] \equiv H_{\text{CM}} + H_{\text{rel}} \quad (10.95)$$

where the reduced mass μ and the total mass M are

$$\mu = \frac{m_1 m_2}{m_1 + m_2}, \quad M = m_1 + m_2 \quad (10.96)$$

Equation (10.95) represents the desired separation of H into the Hamiltonian of the center of mass, H_{CM} , and the Hamiltonian of the coordinates relative to the center of mass, H_{rel} (Fig. 10.9). Since $\mathcal{R}$ is absent in H (i.e., $\mathcal{R}$ is a cyclic coordinate; see Section 1.2), the momentum of the center of mass, $\mathcal{P}$, is constant. The center of mass moves in straight rectilinear motion, characteristic of a free particle of mass M . The motion relative to the center of mass is that of a particle of mass μ moving in the central potential $V(r)$.

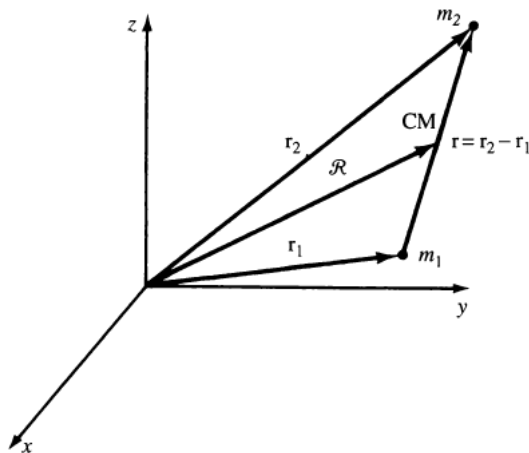


FIGURE 10.9 The relative vector $\mathbf{r}$ and the center-of-mass vector $\mathcal{R}$. In the classical motion, $\mathcal{P} = M\mathcal{R} = \text{constant} = \mathcal{P}(0) = \text{the initial value of } \mathcal{P}(t)$. At any time t

$$\mathcal{R}(t) = \mathcal{R}(0) + \frac{\mathcal{P}(0)t}{M}$$

Solving the dynamical equations (viz., Hamilton's equations) using H_{rel} gives $\mathbf{r}(t)$, which when affixed to $\mathcal{R}(t)$ gives the motion in the "lab frame."

The Transformation of $\hat{H}$

For the quantum mechanical case, the transformation of $\hat{H}$ is again effected with the equations (10.94), which are now interpreted as operator relations. Cartesian components of “old” coordinate and momentum operators ($\hat{\mathbf{r}}_1, \hat{\mathbf{p}}_1; \hat{\mathbf{r}}_2, \hat{\mathbf{p}}_2$) obey the commutation relations

$$\begin{aligned} [\hat{r}_{1j}, \hat{p}_{1j}] &= i\hbar \\ [\hat{r}_{2j}, \hat{p}_{2j}] &= i\hbar \end{aligned} \quad j = 1, 2, 3 \quad (10.97)$$

These are the only nonvanishing commutators. With these relations and (10.94), one obtains that the only nonvanishing commutator relations for components of the “new” operators ($\hat{\mathbf{r}}, \hat{\mathbf{p}}; \hat{\mathcal{R}}, \hat{\mathcal{P}}$) are

$$\begin{aligned} [\hat{r}_j, \hat{p}_j] &= i\hbar \\ [\hat{\mathcal{R}}_j, \hat{\mathcal{P}}_j] &= i\hbar \end{aligned} \quad j = 1, 2, 3 \quad (10.98)$$

Thus, in obtaining

$$\begin{aligned} \hat{H} &= \hat{H}_{\text{CM}} + \hat{H}_{\text{rel}} \\ \hat{H}_{\text{CM}} &= \frac{\hat{\mathcal{P}}^2}{2M} \\ \hat{H}_{\text{rel}} &= \frac{\hat{\mathcal{P}}^2}{2\mu} + V(r) \end{aligned} \quad (10.99)$$

the Hamiltonian is separated into two parts involving components that are independent of one another. For such cases, the Schrödinger equation has product eigenfunctions

$$\tilde{\varphi} = \varphi_{\text{CM}}(\mathcal{R})\varphi_{\text{rel}}(\mathbf{r}) \quad (10.100)$$

and summational eigenvalues

$$\tilde{E} = E_{\text{CM}} + E_{\text{rel}} \quad (10.101)$$

where

$$\begin{aligned} \hat{H}\tilde{\varphi} &= \tilde{E}\tilde{\varphi} \\ \hat{H}_{\text{CM}}\varphi_{\text{CM}} &= E_{\text{CM}}\varphi_{\text{CM}} \\ \hat{H}_{\text{rel}}\varphi_{\text{rel}} &= E_{\text{rel}}\varphi_{\text{rel}} \end{aligned} \quad (10.102)$$

The Schrödinger equation for the center of mass appears explicitly as

$$\frac{\hat{\mathcal{P}}^2}{2M}\varphi_{\text{CM}} = E_{\text{CM}}\varphi_{\text{CM}} \quad (10.103)$$

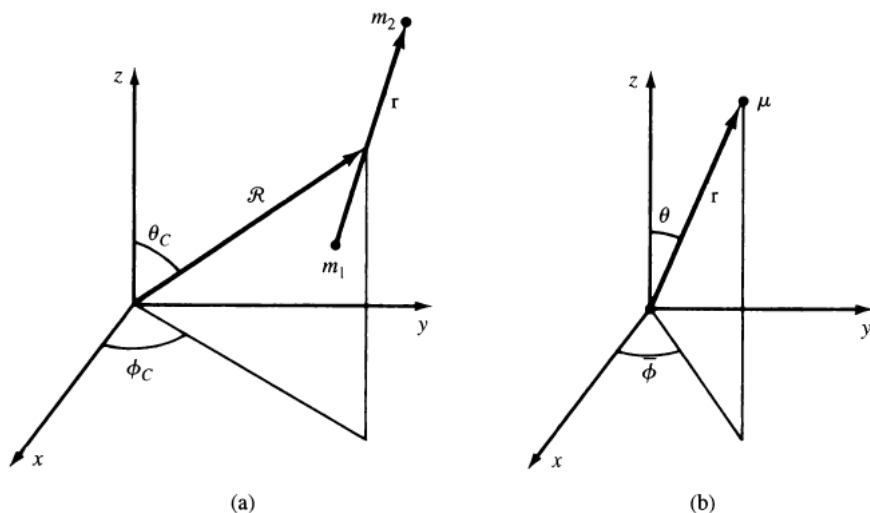


FIGURE 10.10 (a) Spherical angle variables for the center-of-mass radius vector $\mathbf{R}$. (b) Coordinates relative to the center of mass.

This is the Schrödinger equation for a free particle of mass M . Its solution was obtained in the previous section. With the linear momentum $\mathbf{P}$ specified, the states are

$$\begin{aligned}\varphi_{\text{CM}} &= A e^{i\mathbf{K}\cdot\mathbf{R}} \\ \mathbf{P} &= \hbar\mathbf{K}, \quad E_{\text{CM}} = \frac{\hbar^2 K^2}{2M}\end{aligned}\quad (10.104)$$

In the representation where L_{CM}^2 and L_{CM_z} are specified, the eigenstates are

$$\varphi_{\text{CM}} = j_{l_C}(\mathbf{K}\mathbf{R}) Y_{l_C}^{m_C}(\theta_C, \phi_C) \quad (10.105)$$

The spherical coordinates of $\mathbf{R}$ are (R, θ_C, ϕ_C) (see Fig. 10.10a).

Radial Equation for a Central Potential

The Schrödinger equation for φ_{rel} appears as (dropping the “rel” subscript)

$$\left[\frac{\hat{p}^2}{2\mu} + V(r) \right] \varphi = E\varphi \quad (10.106)$$

For central potential functions $V(r)$, it proves most convenient to express the above Hamiltonian in spherical coordinates. The interparticle radius $\mathbf{r}$ has coordinates (r, θ, ϕ) with the polar axis depicted as lying in the z direction (see Fig. 10.10b).

In these coordinates the Schrödinger equation above becomes

$$\hat{H}\varphi = \left[\frac{\hat{p}_r^2}{2\mu} + \frac{\hat{L}^2}{2\mu r^2} + V(r) \right] \varphi = E\varphi \quad (10.107)$$

First, we note that $\hat{L}^2$ and $\hat{L}_z$ both commute with $\hat{H}$. The remaining components in $\hat{H}$ are all isotropic forms and are therefore unaffected by angular momentum operators. It follows that $\hat{H}$, $\hat{L}^2$, and $\hat{L}_z$ have simultaneous eigenstates. These are given by the product form

$$\varphi = R(r)Y_l^m(\theta, \phi) \quad (10.108)$$

Substituting this solution into the Schrödinger equation above gives the “radial” equation

$$\left[\frac{\hat{p}_r^2}{2\mu} + \frac{\hbar^2 l(l+1)}{2\mu r^2} + V(r) \right] R(r) = E R(r) \quad (10.109)$$

This is an ordinary, second-order, linear differential equation for the radial dependent component of the wavefunction $R(r)$. Since only one variable is involved in (10.109), it is suggestive of one-dimensional motion with the effective potential

$$V_{\text{eff}} = V(r) + \frac{\hbar^2 l(l+1)}{2\mu r^2} \quad (10.110)$$

The second term in this expression is called the “angular momentum barrier.” It becomes infinitely high as $r \rightarrow 0$ and acts as a repulsive core, which for $l > 0$ prevents collapse of the system (see Fig. 10.11).

The normalization of the eigenstates (10.108) is given by the integral

$$\begin{aligned} \langle RY_l^m | RY_l^m \rangle &= \int_0^\infty dr r^2 \int_{4\pi} d\Omega |R(r)Y_l^m(\theta, \phi)|^2 = 1 \\ &= \int_0^\infty r^2 |R(r)|^2 dr = 1 \end{aligned} \quad (10.111)$$

The radial displacement r separates the two particles m_1 and m_2 . If we envision particle m_1 at the origin, then

$$Pr^2 dr d\Omega = |R(r)Y_l^m(\theta, \phi)|^2 r^2 dr d\Omega \quad (10.112)$$

is the probability of finding m_2 in the volume element $r^2 dr d\Omega$ about m_1 (an equally valid statement is obtained with m_1 and m_2 reversed). What is the probability of finding m_2 in a spherical shell of radius between r and $r + dr$, about

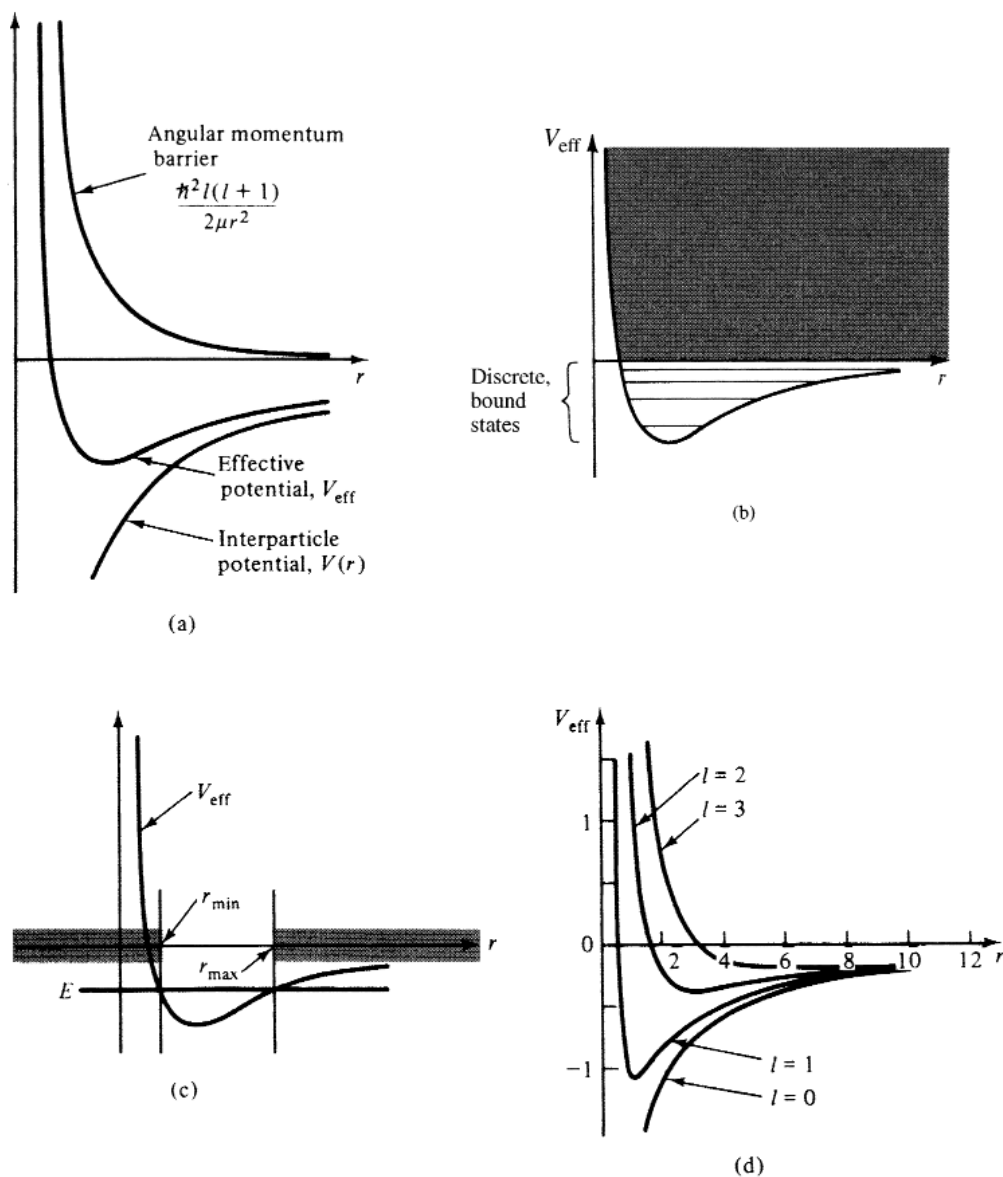


FIGURE 10.11 (a) The effective potential in relation to the angular momentum barrier

$$V_{\text{eff}} = V(r) + \frac{\hbar^2 l(l+1)}{2\mu r^2}$$

(b) Nature of the quantum mechanical spectrum for central potential problems. (c) The classical motion corresponding to the energy E . Shaded regions define classically forbidden domains. (d) The effective potential energy V_{eff} for hydrogen for several values of the orbital quantum number l . Units of r are angstroms. V_{eff} is in units of 10^{-11} erg.

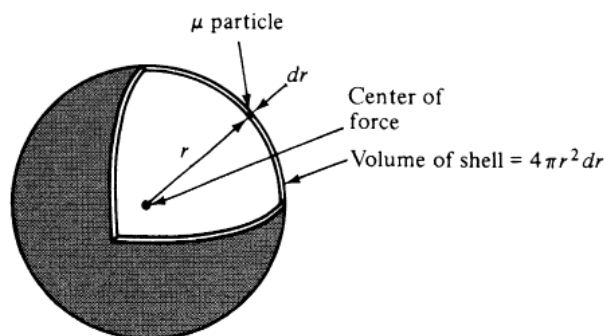


FIGURE 10.12 Probability of finding the fictitious μ particle in a spherical shell between r and $r + dr$ is

$$P_r dr = |R(r)r|^2 dr$$

This is also the probability of finding m_2 in a spherical shell about m_1 in the configuration shown (or m_1 in a shell about m_2).

m_1 ? The answer is (Fig. 10.12)

$$P_r dr = \left(\int_{4\pi} P r^2 d\Omega \right) dr = |R(r)|^2 r^2 dr \equiv |u(r)|^2 dr \quad (10.113)$$

so that

$$\int_0^\infty |u(r)|^2 dr = 1$$

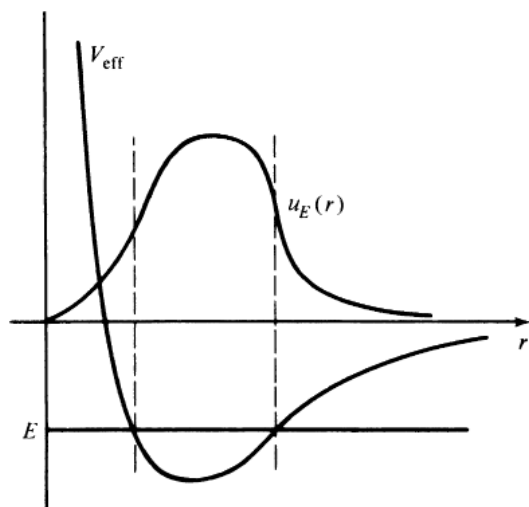


FIGURE 10.13 The radial probability amplitude u_E corresponding to the energy E decays to zero in the classically forbidden domain.

The classically forbidden domains (see Chapter 1) correspond to values of r for which $E < V$. The related property for a spherical quantum mechanical system is that the probability density $|u(r)|^2$ becomes small in these domains (see Figs. 10.11c and 10.13).

Having found the radial function $R(r)$, in a specific two-body problem, the wavefunction for the system relative to the laboratory frame (as opposed to the center-of-mass frame) is either of the forms

$$\begin{aligned}\bar{\varphi} &= Ae^{i\mathcal{P}\cdot\mathcal{R}/\hbar} R(r) Y_l^m(\theta, \phi) \\ \bar{\varphi} &= j_{l_C}(K\mathcal{R}) Y_{l_C}^{m_C}(\theta_C, \phi_C) R(r) Y_l^m(\theta, \phi)\end{aligned}\quad (10.114)$$

In the first representation, the six parameters ($\mathcal{P}$, L^2 , L_z , E) are specified. In the second representation, the six parameters (E_{CM} , L_{CM}^2 , L_{CM_z} ; L^2 , L_z , E) are specified.

Continuity and Boundary Conditions

Some general properties of the radial wavefunction are as follows. With $R(r)$ everywhere bounded, we note first that $u(r) \equiv rR(r)$ must vanish at the origin.¹⁰ For $r > 0$, with energy E and potential energy $V(r)$ bounded, the radial equation (10.109) indicates that

$$\hat{p}_r^2 R(r) = -\hbar^2 \frac{1}{r} \frac{\partial}{\partial r} \left[\frac{\partial}{\partial r} u(r) \right]$$

is likewise bounded. It follows that $\partial u / \partial r$ is continuous. The existence of this derivative implies that $u(r)$ is continuous. The latter two conditions infer continuity of the logarithmic derivative

$$\frac{1}{u} \frac{du}{dr} = \frac{d \ln u}{dr}$$

These conditions on the wavefunction in spherical coordinates are employed in obtaining the ground state of the deuteron (Problem 10.31) and in construction of the bound states of the hydrogen atom as described in the following section. They will also come into play in construction of the states for low-energy scattering from a spherical well (Section 14.2).

PROBLEMS

10.22 Consider a two-particle system. The momenta of the particles are $\mathbf{p}_1$ and $\mathbf{p}_2$, respectively.

¹⁰Dirac obtains this boundary condition from the stipulation that solutions to the Schrödinger equation in spherical coordinates agree with those obtained in Cartesian coordinates. For further discussion, see P. A. M. Dirac, *The Principles of Quantum Mechanics*, 4th ed., Oxford University Press, New York, 1958.

- (a) What is $[\hat{\mathbf{p}}_1, \hat{\mathbf{p}}_2]$? Explain your answer.
- (b) Use the answer to part (a) to show that $[\hat{\mathbf{p}}, \hat{\mathcal{P}}] = 0$, where $\mathbf{p}$ is the “relative” momentum [defined in (10.94)] and $\mathcal{P}$ is the momentum of the center of mass.
- (c) The particles interact under a potential that is a function of the distance between them. As shown in the text, transformation to coordinates relative to the center of mass effects a partitioning of the Hamiltonian, $\hat{H} = \hat{H}_{\text{CM}} + \hat{H}_{\text{rel}}$. What is $[\hat{H}_{\text{CM}}, \hat{H}_{\text{rel}}]$?

10.23 Prove that the following equations are compatible with the transformation equations (10.94).

- (a) $\frac{p_1^2}{2m_1} + \frac{p_2^2}{2m_2} = \frac{p^2}{2\mu} + \frac{\mathcal{P}^2}{2M}$
- (b) $m_1 r_1^2 + m_2 r_2^2 = \mu r^2 + M \mathcal{R}^2$
- (c) $\mathbf{p}_1 \cdot \mathbf{r}_1 + \mathbf{p}_2 \cdot \mathbf{r}_2 = \mathbf{p} \cdot \mathbf{r} + \mathcal{P} \cdot \mathcal{R}$
- (d) $\mathbf{L}_1 + \mathbf{L}_2 = \mathbf{L} + \mathbf{L}_{\text{CM}}$

In part (d),

$$\begin{aligned} \mathbf{L}_1 &= \mathbf{r}_1 \times \mathbf{p}_1 & \mathbf{L} &= \mathbf{r} \times \mathbf{p} \\ \mathbf{L}_2 &= \mathbf{r}_2 \times \mathbf{p}_2 & \mathbf{L}_{\text{CM}} &= \mathcal{R} \times \mathcal{P} \end{aligned}$$

10.24 At a particular time, the wavefunctions of a mass m moving in a three-dimensional potential field is

$$\varphi = A(x + y + z)e^{-k_0 r}$$

- (a) Calculate the normalization constant A .
- (b) What is the probability that measurement of L^2 and L_z finds $2\hbar^2$ and 0, respectively? (See Table 9.1.)
- (c) What is the probability of finding the particle in the sphere $r \leq k_0^{-1}$?
- 10.25** For a two-particle system (m_1, m_2), what is the fractional distance to the center of mass from m_1 and m_2 , respectively? What are these numbers for hydrogen?
- 10.26** Let $\mathbf{e}$ be a unit vector in an arbitrary but fixed direction. Show that the commutators between the components of $\hat{\mathbf{r}}$ and $\hat{\mathbf{p}}$, respectively, with the component $\mathbf{e} \cdot \hat{\mathbf{L}}$ obey the relations

$$\begin{aligned} [\hat{\mathbf{p}}, \mathbf{e} \cdot \hat{\mathbf{L}}] &= i\hbar \mathbf{e} \times \hat{\mathbf{p}} \\ [\hat{\mathbf{r}}, \mathbf{e} \cdot \hat{\mathbf{L}}] &= i\hbar \mathbf{e} \times \hat{\mathbf{r}} \end{aligned}$$

10.27 Use the results of Problem 10.26 to show that $\hat{p}^2$, $\hat{r}^2$, and $\hat{\mathbf{r}} \cdot \hat{\mathbf{p}}$ all commute with every component of $\hat{\mathbf{L}}$. Then show that every component of $\hat{\mathbf{L}}$ commutes with any isotropic function $f(r^2)$.

Answer (partial)

If the statement is true for arbitrary $\mathbf{e}$, it is true for all components of $\hat{\mathbf{L}}$.

$$\begin{aligned} [\hat{p}^2, \mathbf{e} \cdot \hat{\mathbf{L}}] &= \hat{\mathbf{p}}[\hat{\mathbf{p}}, \mathbf{e} \cdot \hat{\mathbf{L}}] + [\hat{\mathbf{p}}, \mathbf{e} \cdot \hat{\mathbf{L}}] \cdot \hat{\mathbf{p}} \\ &= i\hbar(\hat{\mathbf{p}} \cdot \mathbf{e} \times \hat{\mathbf{p}} + \mathbf{e} \times \hat{\mathbf{p}} \cdot \hat{\mathbf{p}}) = 0 \end{aligned}$$

- 10.28** Two free particles of mass m_1 and m_2 , respectively, move in 3-space. They do not interact. Write the eigenfunctions and eigenenergies of this system in as many representations as you can. Indicate the number of parameters specified in the eigenstates associated with these representations.
- 10.29** Write down the time-dependent wavefunction corresponding to eigenstates for a two-particle system in the two representations (10.114).
- 10.30** Consider two particles that attract each other through the potential

$$V(r) = -\frac{\hbar^2 K^2}{2\mu r^2}$$

The displacement between particles is r , K is a constant, and μ is reduced mass. In states of definite angular momentum, what are the values of the angular momentum quantum number l for which the effective force between particles is repulsive?

- 10.31** In Problem 8.1 the depth of the potential well appropriate to a deuteron was evaluated using a one-dimensional approximation. A more refined estimate may be obtained using a three-dimensional spherical well with characteristics

$$V(r) = -|V|, \quad r < a \quad \text{region I}$$

$$V(r) = 0 \quad r \geq a \quad \text{region II}$$

- (a) Construct components of the ground-state wavefunction in regions I and II, respectively.
- (b) Show that matching conditions at $r = a$ give the dispersion relation

$$\eta = -\xi \cot \xi$$

$$\rho^2 = \xi^2 + \eta^2$$

where

$$\rho^2 \equiv \frac{2m|V|a^2}{\hbar^2}, \quad \xi \equiv ka, \quad \eta \equiv \kappa a$$

$$|E| = \frac{\hbar^2 \kappa^2}{2m}, \quad |V| - |E| = \frac{\hbar^2 k^2}{2m}$$

- (c) To within the same approximation suggested in Problem 8.1, obtain a numerical value for the depth $|V|$ of the three-dimensional deuteron well. From the ratio $|E|/|V|$ for this bound state, would you say that the deuteron is a strongly or a weakly bound nucleus?

Answer (partial)

- (a) Component wavefunctions in the well domain, region I, are the spherical Bessel functions. The ground-state component is therefore

$$\varphi_I = \frac{\sin kr}{kr}$$

In region II the component ground-state wavefunction is exponentially damped.

$$\varphi_{II} = A \frac{e^{-\kappa r}}{\kappa r}$$

(b) Here one must invoke continuity of $d \ln u / dr$.

(c) You should obtain the answer $|E|/|V| = 0.08$. The binding energy of a deuteron is 2.2 MeV which, with the preceding estimate, indicates a relatively large well depth. A sketch of the normalized wavefunction further reveals that there is approximately only one chance in three that the nucleons are closer together than the well radius a .

10.32 Show by explicit calculation that the eigenfunctions and eigenenergies as given in Table 10.2 are correct for each of the three respective “box” configurations shown. (Primes denote differentiation.)

10.6 ■ THE HYDROGEN ATOM

Hamiltonian and Eigenenergies

The (relative) Hamiltonian for the hydrogen atom (more accurately, for a “hydrogenic” atom¹¹ of atomic number Z) appears as

$$\hat{H} = \frac{\hat{p}_r^2}{2\mu} + \frac{\hat{L}^2}{2\mu r^2} - \frac{Ze^2}{r} \quad (10.115)$$

The corresponding Schrödinger equation is

$$\left(\frac{\hat{p}_r^2}{2\mu} + \frac{\hat{L}^2}{2\mu r^2} - \frac{Ze^2}{r} \right) \varphi = E\varphi = -|E|\varphi \quad (10.116)$$

We are seeking the *bound states* of hydrogen. These correspond to the negative eigenenergies, $E = -|E|$. Setting $\varphi = R(r)Y_l^m(\theta, \phi)$ in the latter equation gives the radial equation (10.109)

$$\left[\frac{-\hbar^2}{2\mu} \left(\frac{1}{r} \frac{d^2}{dr^2} r \right) + \frac{\hbar^2 l(l+1)}{2\mu r^2} - \frac{Ze^2}{r} + |E| \right] R = 0 \quad (10.117)$$

Changing the dependent variable to

$$u = rR$$

introduced previously in (10.113), gives

$$\left(-\frac{d^2}{dr^2} + \frac{l(l+1)}{r^2} - \frac{2\mu}{\hbar^2} \frac{Ze^2}{r} + \frac{2\mu|E|}{\hbar^2} \right) u = 0 \quad (10.118)$$

¹¹Hydrogenic atoms are atoms that are ionized with all but one electron bound to the nucleus which carries the charge $+Ze$ (e.g., He^+ , Li^{++} , etc.).

Introducing the notation

$$\begin{aligned}\rho &\equiv 2\kappa r, & \frac{\hbar^2 \kappa^2}{2\mu} &= |E| \\ \lambda^2 &= \left(\frac{Z}{\kappa a_0} \right)^2 = \frac{Z^2 \mathbb{R}}{|E|} \\ \mathbb{R} &= \frac{\hbar^2}{2\mu a_0^2}, & a_0 &= \frac{\hbar^2}{\mu e^2}\end{aligned}\tag{10.119}$$

where $\mathbb{R}$ is the Rydberg constant¹² (2.13) and a_0 is the Bohr radius (2.14), the radial equation may be further simplified to the form

$$\frac{d^2 u}{d\rho^2} - \frac{l(l+1)}{\rho^2} u + \left(\frac{\lambda}{\rho} - \frac{1}{4} \right) u = 0\tag{10.120}$$

For large values of ρ this equation reduces to

$$\frac{d^2 u}{d\rho^2} - \frac{u}{4} = 0$$

so that

$$u \sim A e^{-\rho/2} + B e^{\rho/2}$$

In order that u vanish as $\rho \rightarrow \infty$, we set $B = 0$, so

$$u \sim e^{-\rho/2} \quad (\rho \rightarrow \infty)$$

In the neighborhood of the origin, (10.120) reduces to

$$\frac{d^2 u}{d\rho^2} - \frac{l(l+1)}{\rho^2} u = 0$$

Substitution of the trial solution $u = \rho^q$ gives

$$u \sim A \rho^{-l} + B \rho^{l+1}$$

In order for u to vanish at the origin, we must set $A = 0$. This gives

$$u \sim \rho^{l+1} \quad (\rho \rightarrow 0)$$

With these two asymptotic forms at hand, we are prepared to solve (10.120) through a polynomial expansion. Solution in the form

$$\begin{aligned}u(\rho) &= e^{-\rho/2} \rho^{l+1} F(\rho) \\ F(\rho) &= \sum_{i=0}^{\infty} C_i \rho^i\end{aligned}\tag{10.121}$$

¹²The Rydberg constant written with m in place of μ (i.e., assuming infinite proton mass) is sometimes written $\mathbb{R}_{\infty}$.

with F finite everywhere, gives the proper behavior at $\rho \sim 0$ and $\rho \sim \infty$. Substituting (10.121) for u into (10.120), we obtain

$$\left[\rho \frac{d^2}{d\rho^2} + (2l + 2 - \rho) \frac{d}{d\rho} - (l + 1 - \lambda) \right] F(\rho) = 0 \quad (10.122)$$

Note that, for a given value of the orbital quantum number l , this is an eigenvalue equation with eigenvalue λ . The values of λ (or, equivalently, the eigenenergies, $|E|$) are those values which ensure that $F(\rho)$ is finite for all ρ . Substituting the series (10.121) into the latter equation and equating coefficients of equal powers in ρ gives the *recurrence relation*

$$C_{i+1} = \frac{(i + l + 1) - \lambda}{(i + 1)(i + 2l + 2)} C_i \equiv \Gamma_{il} C_i \quad (10.123)$$

In the limit that $i \rightarrow \infty$, this relation becomes

$$C_{i+1} \sim \frac{C_i}{i}$$

which is the same ratio of coefficients obtained in the expansion

$$e^\rho = \sum C_i \rho^i = \sum \frac{\rho^i}{i!}$$

$$\frac{C_{i+1}}{C_i} = \frac{i!}{(i+1)!} = \frac{1}{i+1} \sim \frac{1}{i}$$

It follows that the form of $u(\rho)$ generated by the series (10.121) behaves as

$$u(\rho) \sim e^{-\rho/2} \rho^{l+1} e^\rho = e^{\rho/2} \rho^{l+1}$$

which diverges for large ρ . To obtain a finite wavefunction, the expansion (10.121) for any given value of l must terminate at some finite value of i , which we will call $i_{\max}$. At this value of i , $\Gamma_{il} = 0$. Since all parameters in (10.123) are positive, Γ_{il} can only vanish if

$$i_{\max} + l + 1 = \lambda$$

The function u so generated is a polynomial and, due to the exponential term in the form (10.121), we see that, as demanded, the wavefunction is finite everywhere.

Since i and l are integers, it follows that λ is also an integer, which is called the *principal quantum number*, n .

$$n = i_{\max} + l + 1$$

Thus the above cutoff condition on the series (10.121), which ensures that $u(\rho)$ is finite for all ρ , also serves to determine the eigenenergies λ .

$$\lambda_n^2 = n^2 = \frac{Z^2 \mathbb{R}}{|E_n|}$$

$$E_n = -|E_n| = -\frac{Z^2 \mathbb{R}}{n^2} \quad (10.124)$$

These are the same values found previously in the simpler Bohr model (Section 2.4).

Laguerre Polynomials

The hydrogen eigenfunction corresponding to the eigenvalue E_n is given by (10.121) with the series over i cut off at the value

$$i_{\max} = n - l - 1 \quad (10.125)$$

and the recurrence relation for the coefficients $\{C_i\}$ given by (10.123).

$$u_{nl}(\rho) = e^{-\rho/2} \rho^{l+1} F_{nl}(\rho) = A_{nl} e^{-\rho/2} \rho^{l+1} \sum_{i=0}^{n-l-1} C_i \rho^i \quad (10.126)$$

$$C_{i+1} = \Gamma_{il} C_i, \quad \rho \equiv 2\kappa_n r, \quad \kappa_n = \frac{Z}{a_0 n}$$

where A_{nl} is a normalization constant. The polynomials $F_{nl}(\rho)$ (of order $n-l-1$) so obtained are better known as the *associated Laguerre polynomials*, L_{n-l-1}^{2l+1} (see Table 10.3). The reader should take note of the fact that the scale length of ρ changes with different values of n . This is because the radial displacement r is nondimensionalized through the wavenumber κ_n , which is dependent on n . Note also that eigenstates in Table 10.3 are in terms of $R(r) = u(r)/r$.

Degeneracy

Since $i_{\max} \geq 0$, with (10.125) we obtain

$$l \leq n - 1$$

So for a given value of the *principal quantum number* n , the *orbital quantum number* l cannot exceed the value

$$l_{\max} = n - 1 \quad (10.127)$$

This corresponds to the values $l = 0, 1, 2, \dots, (n - 1)$. Each of these l values corresponds to different values of $i_{\max}$ and therefore different wavefunctions. Inasmuch as the eigenenergy E_n depends only on the principal quantum number n , these n distinct orbital states are degenerate. For instance, there are three distinct radial functions that correspond to the eigenenergy E_3 . These are $u_{3,0}$, $u_{3,1}$, and $u_{3,2}$.

The complete eigenstate of the Hamiltonian (10.115) contains the factor $Y_l^m(\theta, \phi)$ [see (10.108)]. For each value of l , there are $2l + 1$ values of m_l : $m_l = -l, \dots, +l$, which correspond to distinct Y_l^m functions that give the same

TABLE 10.3 Eigenfunctions of hydrogen in terms of associated Laguerre polynomials

The Normalized Eigenfunctions of Hydrogen ($Z = 1$)

$$\varphi_{nlm}(r, \theta, \phi) = (2\kappa)^{3/2} A_{nl} \rho^l e^{-\rho/2} F_{nl}(\rho) Y_l^m(\theta, \phi) = R_{nl}(r) Y_l^m(\theta, \phi)$$

$$\rho = 2\kappa r = \frac{2Z}{a_0 n} r \quad \int_0^\infty |R_{nl}(r)|^2 r^2 dr = 1$$

$$A_{nl} = \sqrt{\frac{(n-l-1)!}{2n[(n+l)!]^3}}$$

$$\varphi_{100} = \frac{1}{\sqrt{8\pi}} \left(\frac{2Z}{a_0}\right)^{3/2} e^{-(Z/a_0)r}$$

$$F_{nl}(\rho) = L_{n-l-1}^{2l+1}(\rho) = L_{i_{\max}}^{2l+1}(\rho) = \sum_{i=0}^{n-l-1} \frac{(-1)^i [(n+l)!]^2 \rho^i}{i! (n-l-1-i)! (2l+1+i)!}$$

Associated Laguerre Polynomials $L_p^q(\rho)$ and Laguerre Polynomials $L_p(\rho)$

Differential equation

$$\left[\rho \frac{d^2}{d\rho^2} + (q+1-\rho) \frac{d}{d\rho} + p \right] L_p^q(\rho) = 0$$

Generating function

$$\frac{e^{-\rho s/(1-s)}}{(1-s)^{q+1}} = \sum_{p=0}^{\infty} \frac{s^p}{(p+q)!} L_p^q(\rho) \quad L_0^p(0) = p!$$

Orthonormality

$$\int_0^\infty e^{-\rho} \rho^q L_p^q L_{p'}^q d\rho = \frac{[(p+q)!]^3}{p!} \delta_{pp'} \quad ,$$

Rodrigues's formula

$$L_p(\rho) \equiv L_p^0(\rho) = e^\rho \frac{d^p}{d\rho^p} (\rho^p e^{-\rho}), \quad L_1(\rho) = 1 - \rho, \quad L_2(\rho) = 2! \left(1 - 2\rho + \frac{\rho^2}{2} \right)$$

$$L_p^q(\rho) \equiv (-1)^q \frac{d^q}{d\rho^q} [L_{q+p}(\rho)]$$

Recurrence relations

$$\rho L_p^q(\rho) = (2p+q+1) L_p^q(\rho) - [(p+1)/(p+q+1)] L_{p+1}^q(\rho) - (p+q)^2 L_{p-1}^q(\rho)$$

$$\left(\rho \frac{d}{d\rho} + q - \rho \right) L_p^q(\rho) = (p+1) L_{p+1}^{q-1}(\rho)$$

$$\frac{d}{d\rho} L_p^q(\rho) = -L_{p-1}^{q+1}(\rho)$$

Relation to Other Notations

Alternative notations for the polynomial L_p^q may be found in other texts. The relation between this notation (B) and our own (A) is given by the following table.^a

TABLE 10.3 (Continued)

Notation A (e.g., found in Merzbacher, Messiah, and here)	Notation B (e.g., found in Pauling and Wilson, Schiff, and Tomonaga)
$(-)^q L_p^q$	L_{p+q}^q
$\rho(L_p^q)'' + (q+1-\rho)(L_p^q)' + pL_p^q = 0$	$\rho(L_{p+q}^q)'' + (q+1-\rho)(L_{p+q}^q)' + pL_{p+q}^q = 0$
$R_{nl} = A(2\kappa)^{3/2} e^{-\rho/2} \rho^l L_{n-l-1}^{2l+1}(\rho)$	$R_{nl} = -A(2\kappa)^{3/2} e^{-\rho/2} \rho^l L_{n+l}^{2l+1}(\rho)$
L_p^q is a polynomial of order p	L_{p+q}^q is a polynomial of order p or, equivalently, L_b^a is a polynomial of order $(b-a)$

The first row in this table tells us that L_p^q is written L_{p+q}^q in notation B. The second row indicates that both L functions satisfy the same differential equation. The third row gives the forms of the radial solution R_{nl} in both notations. Still another notation appears in I. S. Gradshteyn and I. M. Ryzhik,^b where

$$L_p^q(\rho)[\text{here}] = (p+q)! L_p^q(\rho)[G \text{ and } R]$$

^aE. Merzbacher, *Quantum Mechanics*, 2d ed., Wiley, New York, 1970.

A. Messiah, *Quantum Mechanics*, Wiley, New York, 1966.

L. Pauling and E. B. Wilson, *Introduction to Quantum Mechanics*, McGraw-Hill, New York, 1935.

L. Schiff, *Quantum Mechanics*, 3d ed., McGraw-Hill, New York, 1968.

S. Tomonaga, *Quantum Mechanics*, North-Holland, Amsterdam, 1966.

^bI. S. Gradshteyn and I. M. Ryzhik, *Tables of Integrals, Series and Products*, Academic Press, New York, 1965.

TABLE 10.4 Allowed values of l and m_l for $n = 1, 2, 3$

n	1	2	3
l	0	0 1	0 1 2
Spectroscopic notation of state	1S	2S 2P	3S 3P 3D
m_l	0	0 -1, 0, +1	0 -1, 0, +1 -2, -1, 0, +1, +2
Degeneracy of state (n^2)	1	4	9

eigenvalues of $\hat{L}^2$ [i.e., $\hbar^2 l(l+1)$]. All these $2l+1$ functions when substituted into (10.116) give the same radial equation, (10.117), which contains only the orbital number l . It follows that for each solution u_{nl} of (10.120), there are $(2l+1)$ solutions to the Schrödinger equation (10.116) corresponding to the same eigenenergy E_n (see Table 10.4). In this manner we obtain¹³

$$\text{Degeneracy of } E_n = \sum_{l=0}^{n-1} (2l+1) = n^2 \quad (10.128)$$

¹³Including spin, degeneracy of states is $2n^2$. This topic is more fully discussed in Section 12.4.

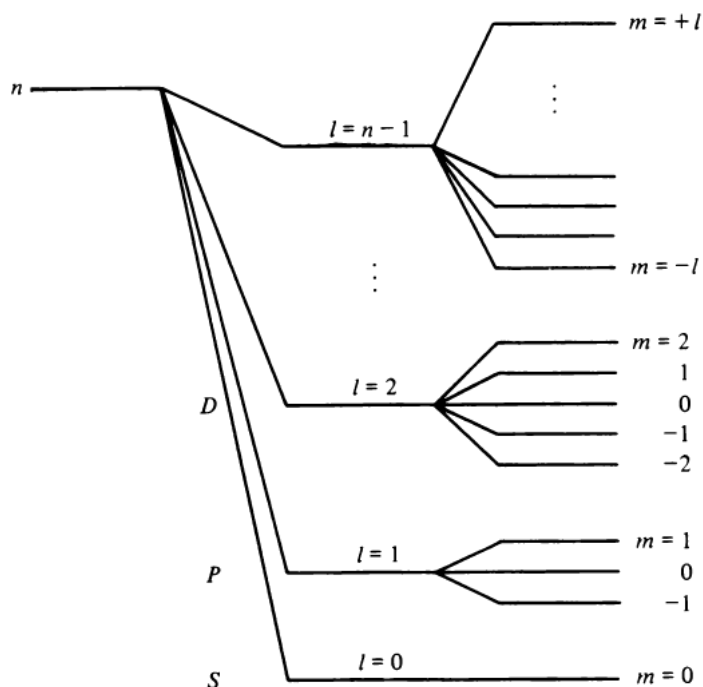


FIGURE 10.14 Term diagram for a hydrogenic atom illustrating all n^2 degenerate states corresponding to the principal quantum number n .

To recapitulate, the allowed values of n , l , and m are (see Fig. 10.14)

$$\begin{aligned}
 n &= 1, 2, 3, \dots \\
 l &= 0, 1, 2, \dots, (n-1) \\
 m &= -l, -l+1, \dots, 0, 1, 2, \dots, +l
 \end{aligned}
 \tag{10.129}$$

Additional Properties of the Eigenstates

The eigenfunctions and eigenenergies of the hydrogenic Hamiltonian (10.115) are

$$\begin{aligned}
 \varphi_{nlm}(r, \theta, \phi) &= R_{nl}(r)Y_l^m(\theta, \phi) \\
 R_{nl} &= \frac{A_{nl}u_{nl}}{r} \\
 E_n &= -\frac{Z^2\mathbb{R}}{n^2} = -\frac{\mu(Ze^2)^2}{2\hbar^2n^2}
 \end{aligned}
 \tag{10.130}$$

A term diagram of these energies is given in Fig. 10.15 (compare Fig. 2.8). The normalization constant A_{nl} (see Table 10.3) is determined by the condition

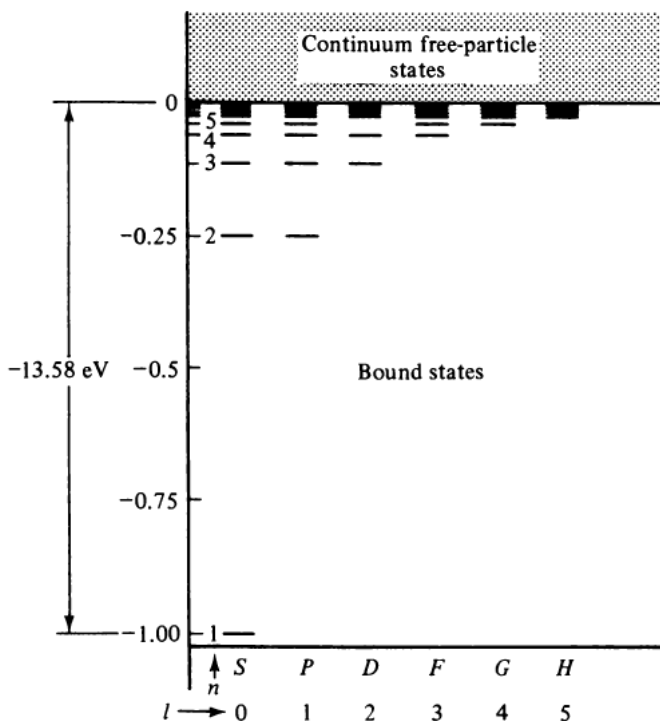


FIGURE 10.15 Energy-level diagram for hydrogen, including the $l \leq 5$ terms. Energy is measured in units of $\mathcal{R}$.

$$\begin{aligned}\langle \varphi_{nlm} | \varphi_{nlm} \rangle &= \int_{4\pi} d\Omega \int_0^\infty r^2 dr \varphi_{nlm}^* \varphi_{nlm} \\ &= |A_{nl}|^2 \int_0^\infty |u_{nl}|^2 dr = 1\end{aligned}$$

Note also the orthogonality of these functions

$$\langle \varphi_{n'l'm'} | \varphi_{nlm} \rangle = \delta_{nn'} \delta_{ll'} \delta_{mm'}$$

The Ground State

To construct the *ground state* φ_{100} ($n = 1, l = 0, m = 0$) we must first find u_{10} . From (10.126), with $C_0 = 1$, and inserting the normalization constant¹⁴ A_{10} , one obtains

$$u_{10} = A_{10} e^{-\rho/2} \rho$$

¹⁴Alternatively, we may take $C_0 = A_{10}$.

Normalization gives

$$A_{10}^2 \frac{1}{2\kappa_1} \int_0^\infty \rho^2 e^{-\rho} d\rho = 1$$

$$A_{10}^2 = \frac{1}{a_0}$$

This gives the normalized ground-state wavefunction (with $Z = 1$)

$$u_{10} = \frac{1}{\sqrt{a_0}} \rho e^{-\rho/2} = \frac{2r}{a_0^{3/2}} e^{-r/a_0}$$

$$R_{10} = \frac{u_{10}}{r} = \frac{2}{a_0^{3/2}} e^{-r/a_0}$$

$$\varphi_{100} = R_{10} Y_0^0 = \frac{2}{(4\pi)^{1/2} a_0^{3/2}} e^{-r/a_0}$$

Additional Properties

The first few normalized eigenstates of hydrogen, with corresponding eigenenergies obtained as outlined above, are listed in Table 10.5.

In Fig. 10.16 nondimensionalized radial functions, $\bar{R}_{nl} = a_0^{3/2} R_{nl}$, are plotted together with the corresponding nondimensionalized probability density functions, $\bar{P}_{nl} = 4\pi a_0 u_{nl}^2 = 4\pi a_0 P_r$. These sketches reveal the shell structure of hydrogen found earlier in the Bohr theory.

The time development of the states of hydrogen follows from (3.70). Consider that

$$\psi(\mathbf{r}, 0) = \varphi_{nlm} = R_{nl} Y_l^m \quad (10.131)$$

The state at time $t \geq 0$ is then

$$\psi(\mathbf{r}, t) = e^{-i\hat{H}t/\hbar} \psi(\mathbf{r}, 0) = e^{-iE_n t/\hbar} \varphi_{nlm} \quad (10.132)$$

The charge density associated with this state is

$$q(\mathbf{r}, t) = e|\psi_{nlm}(\mathbf{r}, t)|^2 = q(\mathbf{r}) = e|\varphi_{nlm}(\mathbf{r})|^2 \quad (10.133)$$

which is independent of time. The electronic charge is e . Thus the atom suffers no radiation in these states. This topic will be returned to in the next section.

The density configurations, $|\varphi_{nlm}|^2$, corresponding to some of the eigenstates of hydrogen are sketched in Fig. 10.17. Since the angular dependence of $|\varphi_{nlm}|^2$ is entirely contained in the factor $|Y_l^m|^2$, it follows that $|\varphi_{nlm}|^2$ is independent of the azimuthal angle ϕ [see (9.78)]. It is rotationally symmetric about the z axis. Thus we need only present a representation of $|\varphi|^2$ in any plane which includes the z axis, such as is depicted in Fig. 10.17. The value of $|\varphi|^2$ is proportional to the density of whiteness in each of the states depicted.

TABLE 10.5 Several normalized time-independent eigenstates of hydrogen

Spectroscopic Notation	Normalized Time-Independent Eigenstates
1S	$\varphi_{100} = \frac{2}{a_0^{3/2}} e^{-r/a_0} Y_0^0(\theta, \phi)$
2S	$\varphi_{200} = \frac{2}{(2a_0)^{3/2}} (1 - r/2a_0) e^{-r/2a_0} Y_0^0(\theta, \phi)$
2P	$\begin{pmatrix} \varphi_{211} \\ \varphi_{210} \\ \varphi_{21-1} \end{pmatrix} = \frac{1}{\sqrt{3} (2a_0)^{3/2}} \frac{r}{a_0} e^{-r/2a_0} \begin{pmatrix} Y_1^1(\theta, \phi) \\ Y_1^0(\theta, \phi) \\ Y_1^{-1}(\theta, \phi) \end{pmatrix}$
3S	$\varphi_{300} = \frac{2}{3(3a_0)^{3/2}} [3 - 2r/a_0 + 2(r/3a_0)^2] e^{-r/3a_0} Y_0^0(\theta, \phi)$
3P	$\begin{pmatrix} \varphi_{311} \\ \varphi_{310} \\ \varphi_{31-1} \end{pmatrix} = \frac{4\sqrt{2}}{9(3a_0)^{3/2}} \frac{r}{a_0} (1 - r/6a_0) e^{-r/3a_0} \begin{pmatrix} Y_1^1(\theta, \phi) \\ Y_1^0(\theta, \phi) \\ Y_1^{-1}(\theta, \phi) \end{pmatrix}$
3D	$\begin{pmatrix} \varphi_{322} \\ \varphi_{321} \\ \varphi_{320} \\ \varphi_{32-1} \\ \varphi_{32-2} \end{pmatrix} = \frac{2\sqrt{2}}{27\sqrt{5} (3a_0)^{3/2}} \left(\frac{r}{a_0}\right)^2 e^{-r/3a_0} \begin{pmatrix} Y_2^2(\theta, \phi) \\ Y_2^1(\theta, \phi) \\ Y_2^0(\theta, \phi) \\ Y_2^{-1}(\theta, \phi) \\ Y_2^{-2}(\theta, \phi) \end{pmatrix}$

Hybridization

Having discovered the wavefunctions of hydrogen, a significant consequence of quantum mechanics emerges important to the formation of molecules. This property concerns the manner in which atomic wavefunctions exhibit geometric orientation in atom–atom binding.

Consider the 2P states of hydrogen listed in Table 10.5. Owing to their degeneracy, any linear combination of these states is an eigenstate corresponding to the same eigenenergy, $-\mathbb{R}/4$. Thus, for example, the following set of three orthogonal wavefunctions spans the same subdimensional Hilbert space as the original three 2P states:

$$\varphi_{2P_x} = \frac{1}{\sqrt{2}}(\varphi_{211} - \varphi_{21-1}) = A x e^{-r/2a_0} \quad (10.134a)$$

$$\varphi_{2P_y} = \frac{1}{\sqrt{2}}(\varphi_{211} + \varphi_{21-1}) = -i A y e^{-r/2a_0} \quad (10.134b)$$

$$\varphi_{2P_z} = \varphi_{210} = A z e^{-r/2a_0} \quad (10.134c)$$

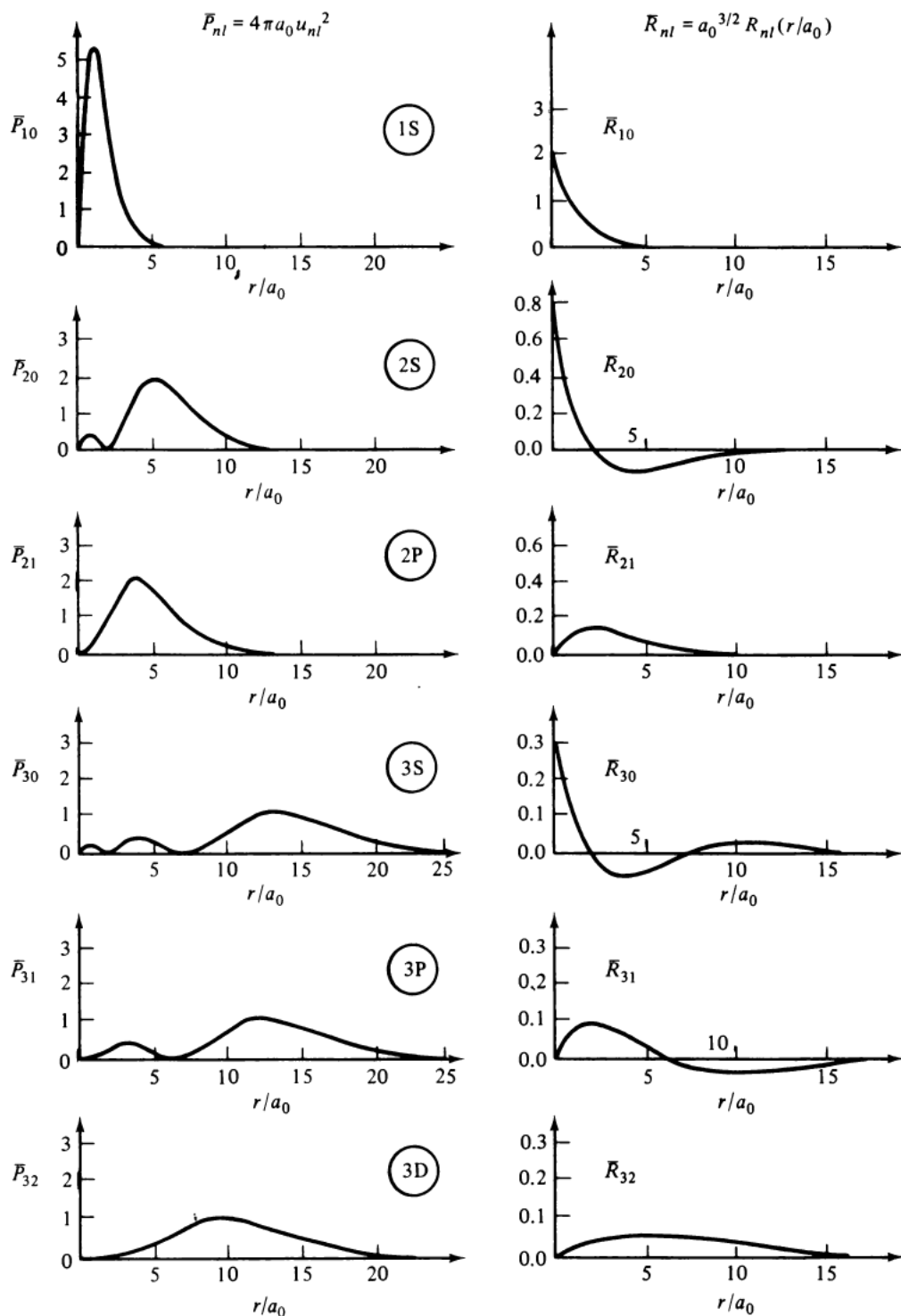


FIGURE 10.16 Nondimensional radial probability density $\bar{P}$ and nondimensional radial wavefunction $\bar{R}$, versus nondimensional radius r/a_0 , for hydrogen. Note that the probability density $\bar{P}$ exhibits the shell structure of the atom.

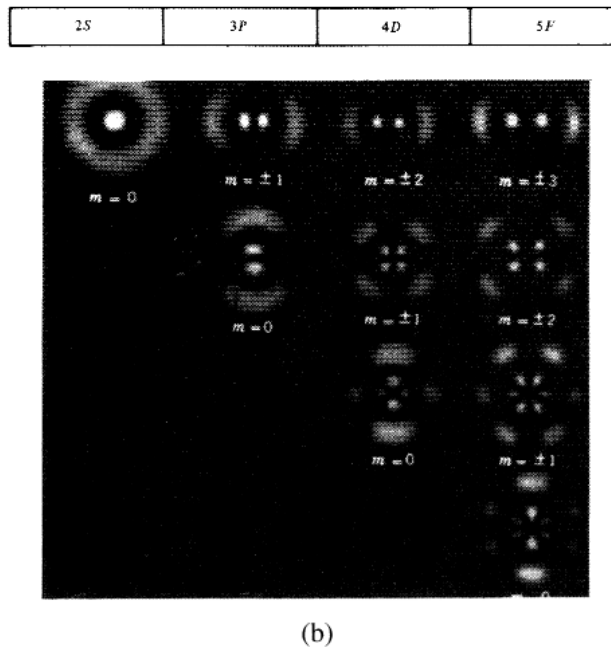
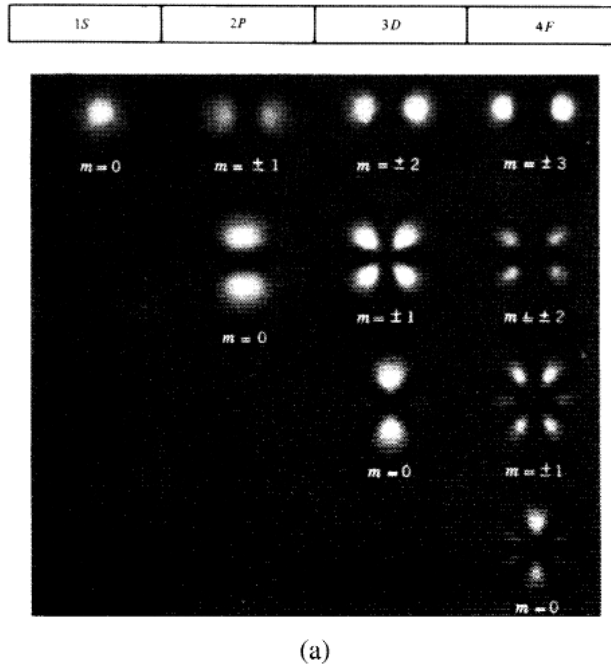


FIGURE 10.17 Probability density, $|\varphi_{nlm}|^2$, for various states of hydrogen. The plane of the paper contains the polar axis, which points from the bottom to the top of the figure. From *Principles of Modern Physics*, by R. B. Leighton. Copyright 1959 by McGraw-Hill. Used with permission of the McGraw-Hill Company.

Angular plots (at fixed r) of the corresponding probability densities $|\varphi_{2P_x}|^2$, $|\varphi_{2P_y}|^2$, $|\varphi_{2P_z}|^2$, reveal that they are figure-eight surfaces of revolution about the x , y , and z axes, respectively (similar to those shown in Fig. 9.10).

Linear combinations of the wavefunctions (10.134) come into play in the formation of the methane molecule CH_4 . The four outer-shell electrons of carbon are in the $2s^2 2p^2$ configuration (see Table 12.2). Wavefunctions appropriate to these four electrons are formed from linear combinations of the three $2P$ states (10.134) together with the $2S$ state of hydrogen and are given by

$$\psi_1 = \frac{1}{\sqrt{4}}(\varphi_{2S} + \varphi_{2P_x} + \varphi_{2P_y} + \varphi_{2P_z}) \quad (10.135a)$$

$$\psi_2 = \frac{1}{\sqrt{4}}(\varphi_{2S} + \varphi_{2P_x} - \varphi_{2P_y} + \varphi_{2P_z}) \quad (10.135b)$$

$$\psi_3 = \frac{1}{\sqrt{4}}(\varphi_{2S} + \varphi_{2P_x} - \varphi_{2P_y} - \varphi_{2P_z}) \quad (10.135c)$$

$$\psi_4 = \frac{1}{\sqrt{4}}(\varphi_{2S} - \varphi_{2P_x} + \varphi_{2P_y} - \varphi_{2P_z}) \quad (10.135d)$$

With the four outer electrons of carbon in these respective four ψ orbitals, the following picture emerges. Angular plots of the probability densities of these wavefunctions about a common origin reveal maxima along the $(1, 1, 1)$,

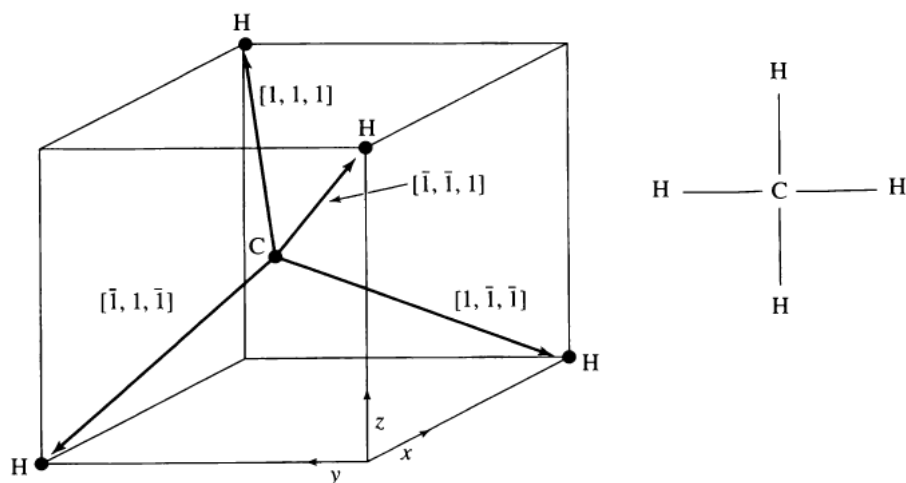


FIGURE 10.18 Orientation of C and H atoms in the CH_4 molecule. Here we have employed crystallographic notation for direction, where a bar over a direction number indicates negative direction. Shown also is the two-dimensional representation of this molecule. Generalizations of this diagram for $\text{C}_n\text{H}_{2n+2}$ hydrocarbons are evident. (Straight-chain alkanes.)

$(-1, -1, 1)$, $(1, -1, -1)$, and $(-1, 1, -1)$ directions, respectively. With the carbon atom at the center of a cube, these orbitals reach out to four tetrahedral corners and covalently bond with hydrogen atoms at these sites to form the CH_4 molecule. See Fig. 10.18.

These orbitals come into play in the chaining of CH_3 molecules. Thus, for example, in the formation of C_2H_6 , an uncoupled electronic ψ orbital of CH_3 bonds with a like orbital of an adjacent CH_3 molecule to form the ethane molecule. The concept of atomic bonding in the formation of molecules is returned to in Chapter 12.

PROBLEMS

10.33 With $C_0 = 1$ in the recurrence relation (10.123), obtain C_1 . Then use (10.126) to show that

$$u_{20} = A_{20} r e^{-r/2a_0} \left(1 - \frac{r}{2a_0} \right)$$

Calculate A_{20} and φ_{200} . Check your answer with the value given in Table 10.5.

10.34 Show that u_{10}^2 has its maximum at $r = a_0$, the Bohr radius.

10.35 Solve the equation $f_x + f = 0$ by expansion technique and check with the solution $f = Ae^{-x}$.

Answer

Assume that

$$f = \sum_0^{\infty} C_i x^i$$

to obtain

$$\sum_0^{\infty} C_i i x^{i-1} + \sum_0^{\infty} C_i x^i = 0$$

With $s = i - 1$ in the first series, we get

$$\sum_{s=-1}^{\infty} C_{s+1} (s+1) x^s + \sum_{i=0}^{\infty} C_i x^i = 0$$

Since the first term in the first series is zero, we may write this equation in the form

$$\sum_{i=0}^{\infty} [C_{i+1} (i+1) + C_i] x^i = 0$$

which is satisfied if and only if

$$C_{i+1} = -\frac{C_i}{i+1}$$

10.36 The average energy of the hydrogen atom in an arbitrary bound state $\chi(\mathbf{r})$ is given by the integral

$$\langle E \rangle = \langle \chi(\mathbf{r}) | \hat{H} | \chi(\mathbf{r}) \rangle$$

Show that

$$\langle E \rangle \geq \langle 100 | \hat{H} | 100 \rangle = E_1$$

Answer

Since $\chi(\mathbf{r})$ lies in the Hilbert space spanned by the basis $\{\varphi_{nlm}\}$, we may expand

$$|\chi\rangle = \sum_{n=1}^{\infty} \sum_{l=0}^{n-1} \sum_{m=-l}^l b_{nlm} |nlm\rangle$$

so that

$$\begin{aligned} \langle \chi | \hat{H} | \chi \rangle &= \sum_n \sum_l \sum_m |b_{nlm}|^2 E_n \\ \sum_n \sum_l \sum_m |b|^2 &= 1 \end{aligned}$$

(The ket vector $|nlm\rangle$ represents the state φ_{nlm} .) Owing to the fact that all eigenenergies are negative, $E_n \leq 0$, the statement to be proven is equivalent to the inequality

$$\begin{aligned} |\langle E \rangle| &\leq |E_1| \\ |\langle E \rangle| &= \sum_n \sum_l \sum_m |b|^2 |E_n| \\ &\leq |E|_{\max} \sum_n \sum_l \sum_m |b|^2 = |E|_{\max} = |E_1| \end{aligned}$$

10.37 (a) What is the effective Bohr radius and ground-state energy for each of the following two-particle systems?

- (1) H^2 , a deuteron and an electron (heavy hydrogen).
 - (2) He^+ , a singly ionized helium atom.
 - (3) Positronium, a bound positron and electron.
 - (4) Mesonium, a proton and negative μ meson. The μ meson has mass $207m_e$ and lasts $\sim 10^{-6}$ s.
 - (5) Two neutrons bound together by their gravitational field.
- (b) Calculate the frequencies of the $(n=2) \rightarrow (n=1)$ transition for each of the systems above.

10.38 At time $t=0$, a hydrogen atom is in the superposition state

$$\psi(\mathbf{r}, 0) = \frac{4}{(2a_0)^{3/2}} \left[\frac{e^{-r/a_0}}{\sqrt{4\pi}} + A \frac{r}{a_0} e^{-r/2a_0} (-iY_1^1 + Y_1^{-1} + \sqrt{7}Y_1^0) \right]$$

- (a) Calculate the value of the normalization constant A .
- (b) What is the probability that measurement of L^2 finds the value $\hbar^2 l(l+1)$?
- (c) What is the probability density $P_r(r)$ [see (10.113)] that the electron is found in the shell of thickness dr about the proton at the radius r ?

- (d) At what value of r is $P_r(r)$ maximum?
 (e) Given the initial state $\psi(\mathbf{r}, 0)$, what is $\psi(\mathbf{r}, t)$?
 (f) What is $\psi(\mathbf{r}, t)$ if at $t = 0$, measurement of L_z finds the value $\hbar$?
 (g) What is $\psi(\mathbf{r}, t)$ if at $t = 0$, measurement of L_z finds the value zero?
 (h) What is the expectation of the “spherical energy operator,” $\langle H_S \rangle$, where

$$\hat{H}_S \equiv \hat{H} - \frac{\hat{p}_r^2}{2\mu}$$

at $t = 0$?

- (i) What is the lowest value of energy that measurement will find at $t = 0$? (Lowest means the negative value farthest removed from zero.)
- 10.39** Find the lowest energy and the smallest value for the classical turning radius of the H -atom electron in the state $l = 6$ (see Fig. 10.11).
- 10.40** In what sense does the Bohr analysis of the hydrogen atom give erroneous results for the magnitude of angular momentum, L ?

Answer

The Bohr analysis that yields the eigenenergies $-\mathbb{R}/n^2$ assumes circular orbits. Circular orbits do not exist in the Schrödinger theory. Quantization of the action, $\oint p_\theta d\theta$, in the Bohr theory gives $L = n\hbar$. In the Schrödinger theory, the maximum value of L is $\hbar\sqrt{n(n+1)}$, which is less than the value that L assumes ($n\hbar$) in the Bohr theory.

- 10.41** What is the ionization energy of a hydrogen atom in the $3P$ state?
- 10.42** Show that $R_{nl}(r)$ has $(n - l - 1)$ zeros (not counting zeros at $r = 0$ and $r = \infty$).
- 10.43** (a) Show that the expectation of the interaction potential $V(r)$ for hydrogenic atoms is

$$\langle nlm|V(r)|nlm\rangle = -\left\langle \frac{Ze^2}{r} \right\rangle = -\frac{\mu Z^2 e^4}{\hbar^2 n^2} = -\frac{2Z^2 \mathbb{R}}{n^2}$$

- (b) Calculate $\langle nlm|\hat{T}|nlm\rangle$, where the kinetic-energy operator $\hat{T}$ is given by

$$\hat{T} = \frac{\hat{p}_r^2}{2\mu} + \frac{\hat{L}^2}{2\mu r^2}$$

What relation do $\langle T \rangle$ and $\langle V \rangle$ satisfy (*the virial theorem*)?

- 10.44** Obtain an explicit expression for the probability density $P_r(r)$ corresponding to the state whose energy is E_2 , for a hydrogenic atom [see (10.113)].

Answer

There are four degenerate eigenstates corresponding to the energy E_2 . Since no direction is preferred for a Hamiltonian whose only interaction term is the central potential $V(r)$, all these degenerate states carry the same “weight” (all lm states are equally probable). There results

$$\begin{aligned}
 4\pi P(r)r^2 dr &= \int_{4\pi} \times \frac{1}{4} [\varphi_{200}^* \varphi_{200} + \varphi_{21-1}^* \varphi_{21-1} + \varphi_{210}^* \varphi_{210} + \varphi_{211}^* \varphi_{211}] \times r^2 dr d\Omega \\
 &= \int_{4\pi} \frac{1}{128\pi} \left(\frac{Z}{a_0}\right)^3 e^{-Zr/a_0} \left[\left(2 - \frac{Zr}{a_0}\right)^2 + \left(\frac{Zr}{a_0}\right)^2 \right. \\
 &\quad \left. \times \left(\frac{1}{2} \sin^2 \theta + \frac{1}{2} \sin^2 \theta + \cos^2 \theta\right) \right] \times r^2 dr d\Omega \\
 P(r) &= \frac{1}{128\pi} \left(\frac{Z}{a_0}\right)^3 e^{-Zr/a_0} \left[\left(2 - \frac{Zr}{a_0}\right)^2 + \left(\frac{Zr}{a_0}\right)^2 \right], \\
 P_r(r) &= 4\pi r^2 P(r)
 \end{aligned}$$

10.45 Give a physical argument in support of the conjecture that the sum

$$\sum_n \equiv \sum_{l=0}^{n-1} \sum_{m=-l}^{+l} [Y_l^m]^* [Y_l^m] [R_{nl}]^2$$

is independent of θ or ϕ .

10.46 Show that for a hydrogen atom in the state corresponding to maximum orbital angular momentum ($l = n - 1$),

$$\begin{aligned}
 \langle n, n-1 | r | n, n-1 \rangle &= a_0 n \left(n + \frac{1}{2} \right) \\
 \langle n, n-1 | r^2 | n, n-1 \rangle &= a_0^2 n^2 (n+1) \left(n + \frac{1}{2} \right)
 \end{aligned}$$

10.47 Use the result of Problem 10.46 to show that for large values of n and l ,

$$\begin{aligned}
 \sqrt{\langle r^2 \rangle} &\rightarrow a_0 n^2 \\
 \frac{\Delta r}{\langle r \rangle} &\rightarrow 0 \\
 E_n &\rightarrow -\frac{1}{2} \frac{e^2}{n^2 a_0}
 \end{aligned}$$

That is, show that for large values of n , the electron is located near the surface of a sphere of radius $a_0 n^2$ and has energy which is the same as that of a classical electron in a circular orbit of the same radius. Recall that $(\Delta r)^2 = \langle r^2 \rangle - \langle r \rangle^2$.

10.48 Calculate $\langle \mathbf{r} \rangle$ in the state φ_{nlm} of hydrogen.

Answer

$$\begin{aligned}
 \langle \mathbf{r} \rangle &= \mathbf{e}_x \langle x \rangle + \mathbf{e}_y \langle y \rangle + \mathbf{e}_z \langle z \rangle \\
 \langle x \rangle &= \iiint r \cos \phi \sin \theta |Y_l^m|^2 R_{nl}^2 r^2 dr d\cos \theta d\phi \\
 &= 0
 \end{aligned}$$

since

$$\int_0^{2\pi} \cos \phi \, d\phi = 0$$

and $|Y_l^m|^2$ is independent of ϕ . Similarly, $\langle y \rangle = 0$. For $\langle z \rangle$ we must calculate

$$\langle z \rangle = \int_{-1}^1 |P_l^m|^2 \cos \theta \, d \cos \theta \int \dots$$

Using the recurrence relations listed in Table 9.3, we find that $\langle z \rangle = 0$. It follows that $\langle \mathbf{r} \rangle = 0$.

10.49 Establish the following properties for hydrogen in the stationary state φ_{nlm} .

$$(a) \quad \frac{2+1}{n^2} \langle r^s \rangle - (2s+1)a_0 \langle r^{s-1} \rangle + \frac{s}{4} [(2l+1)^2 - s^2] a_0^2 \langle r^{s-2} \rangle = 0, \quad s > -2l-1$$

$$(b) \quad \langle r \rangle = n^2 \left[1 + \frac{1}{2} \left(1 - \frac{l(l+1)}{n^2} \right) \right] a_0$$

$$(c) \quad \left\langle \frac{1}{r^2} \right\rangle = \frac{2}{(2l+1)n^3 a_0^2}$$

$$(d) \quad \langle r^2 \rangle = \frac{1}{2} [5n^2 + 1 - 3l(l+1)] n^2 a_0^2$$

$$(e) \quad \left\langle \frac{1}{r} \right\rangle = \frac{1}{n^2 a_0}$$

$$(f) \quad \left\langle \frac{1}{r^3} \right\rangle = \frac{2}{a_0^3 n^3 l(l+1)(2l+1)}$$

[Hint: Multiply (10.120) by $\{\rho^{s+1}u' + [(s+1)/2]\rho^s u\}$ and integrate by parts several times. Note that for hydrogenic atoms, a_0 is replaced by a_0/Z .]

10.50 Show that the most probable values of r for the $l = n-1$ states of hydrogen are

$$\tilde{r} = n^2 a_0$$

These are values that satisfy the equation

$$\frac{d}{dr} (u_{nl})^2 = 0$$

10.7 ■ ELEMENTARY THEORY OF RADIATION

In the last section we found that the hydrogen atom does not radiate in its eigen (stationary) states. The charge density (10.133) is fixed in space with configurations such as depicted in Fig. 10.17. In these states the hydrogen atom is stable against radiation. This is opposed to the classical description in which the electron loses kinetic energy to the radiation field and collapses to the nucleus (see Section 2.1).

The student may be perplexed about the absence of radiation from the state ψ_{nlm} . He/she may well ask: Doesn't the electron have a well-defined angular momentum in such a state, and doesn't this correspond to accelerated motion which gives rise to radiation? His/her friend answers: Maybe the orbit of the electron is

so peculiar that, on the average, the radiation field washes out. After all, we know that if L^2 is specified, two of the three components of $\mathbf{L}$ remain uncertain.

The best way to see what the electron is doing in quantum mechanics is to calculate $\langle \mathbf{r} \rangle$. Specifically, we must calculate this expectation in the state ψ_{nlm} . Suppose that we find $\langle \mathbf{r} \rangle \sim \mathbf{e}_z \cos \omega t$. Then the electron is suffering linear, simple harmonic oscillation. Such oscillation gives rise to *dipole radiation*. But we have already calculated $\langle \mathbf{r} \rangle$ in Problem 10.48, where we found that $\langle \mathbf{r} \rangle = 0$. Not only is $\langle \mathbf{r} \rangle$ time-independent in the eigenstates of hydrogen, but it is also centered at the origin. Note that we may also reach this conclusion by the much simpler argument: Calculate

$$\langle \mathbf{r} \rangle = \langle \psi_{nlm} | \mathbf{r} | \psi_{nlm} \rangle = \iiint \mathbf{r} |\psi_{nlm}|^2 d\mathbf{r}$$

The average of $\mathbf{r}$ is independent of time; hence it must also be zero since the Hamiltonian, (10.115), is isotropic. It contains no vectors. It in no way implies a “preferred” direction, so $\langle \mathbf{r} \rangle$ cannot be a finite constant vector.

Thus while the stability of the hydrogen atom to radiative collapse is totally inexplicable on classical grounds, our quantum mechanical model renders a denumerably infinite set of states $\{\psi_{nlm}\}$ in which the atom suffers no radiation.

How, then, does the atom radiate? In the Bohr theory of radiation, we recall that a photon is emitted when there is a transition from one eigenenergy state to a lower one. Such a decay might be induced by the collision of the atom with another atom in a gas. It might also be induced by collision with an electron in a discharge tube. It might also be induced by collision with a photon in the interior of a star.¹⁵

Suppose that at time $t = 0$, the atom is in an excited (stationary) state ψ_n (n denotes the sequence nlm). The atom is perturbed, emits radiation, and decays to the state $\psi_{n'}$ (Fig. 10.19). We may conclude that in the interim, the atom is in the superposition state

$$\psi = a\psi_n + b\psi_{n'} \quad |a|^2 + |b|^2 = 1 \quad (10.136)$$

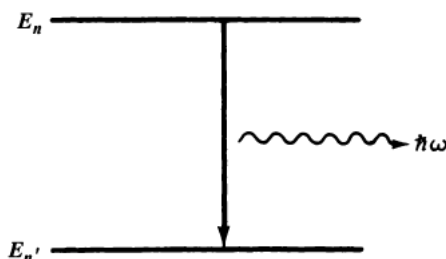


FIGURE 10.19 Atom emits a photon in decaying from state ψ_n to $\psi_{n'}$.

¹⁵Fundamentally, all these collision processes involve the exchange of photons. For further discussion, see E. G. Harris, *A Pedestrian Approach to Quantum Field Theory*, Wiley, New York, 1972.

At any time t , in this interim (by the superposition principle) $|a|^2$ represents the probability that the atom is in the state ψ_n and $|b|^2$ represents the probability that it is in the state $\psi_{n'}$. These coefficients are therefore time-dependent. At $t = 0$, ($|a| = 1, |b| = 0$). At $t = \infty$, ($|a| = 0, |b| = 1$). Let us calculate the expected value of the position of the electron during this collapse.

$$\begin{aligned}\langle \mathbf{r} \rangle &= \langle a\psi_n + b\psi_{n'} | \mathbf{r} | a\psi_n + b\psi_{n'} \rangle \\ &= |a|^2 \langle \psi_n | \mathbf{r} | \psi_n \rangle + |b|^2 \langle \psi_{n'} | \mathbf{r} | \psi_{n'} \rangle \\ &\quad + a^*b \langle \psi_n | \mathbf{r} | \psi_{n'} \rangle + b^*a \langle \psi_{n'} | \mathbf{r} | \psi_n \rangle\end{aligned}$$

The first two terms are time-independent and do not contribute to radiation. The last two terms combine to yield

$$\begin{aligned}\langle \mathbf{r}(t) \rangle &= a^*b e^{i(E_n - E_{n'})t/\hbar} \langle \psi_n | \mathbf{r} | \psi_{n'} \rangle + b^*a e^{i(E_{n'} - E_n)t/\hbar} \langle \psi_{n'} | \mathbf{r} | \psi_n \rangle \quad (10.137) \\ &= 2 \operatorname{Re} [a^*b \langle \psi_n | \mathbf{r} | \psi_{n'} \rangle e^{i(E_n - E_{n'})t/\hbar}] \\ &= 2 |a^*b \langle \psi_n | \mathbf{r} | \psi_{n'} \rangle| \cos(\omega_{nn'}t + \delta) = 2 |\mathbf{r}_{nn'}| \cos(\omega_{nn'}t + \delta)\end{aligned}$$

where $\omega_{nn'}$ is the Bohr frequency

$$\hbar\omega_{nn'} = E_n - E_{n'}$$

δ is a phase factor and $|a^*b|$ is assumed to be slowly varying and of order unity.

Atomic transitions typically occur in an interval of the order of 10^{-9} s. The frequency of emitted radiation, on the other hand, is typically of the order of 10^{15} s^{-1} , so the radiative oscillatory behavior of $\langle \mathbf{r}(t) \rangle$ is due almost exclusively to the cos term, with accompanying Bohr frequency $\omega_{nn'}$ (Fig. 10.20). When the atom is undergoing a transition between the states ψ_n and $\psi_{n'}$, the average position of the electron oscillates with the Bohr frequency corresponding to the energy difference between these states. At the beginning and conclusion of the transition, the atom is in stationary states in which it does not radiate. (These topics are returned to in Section 13.9.)

Selection Rules

Harmonic oscillation of an electron about a proton gives rise to what is commonly referred to as *dipole radiation*¹⁶ (Fig. 10.21). The average radiated power from such an oscillating dipole is¹⁷

$$P = \frac{1}{3} \frac{\omega^4}{c^3} |\mathbf{d}|^2 \quad (10.138)$$

¹⁶An atom also radiates in higher multipole channels (e.g., quadrupole). For the most part, dipole radiation is predominant.

¹⁷See J. D. Jackson, *Classical Electrodynamics*, 2d ed., Wiley, New York, 1975.

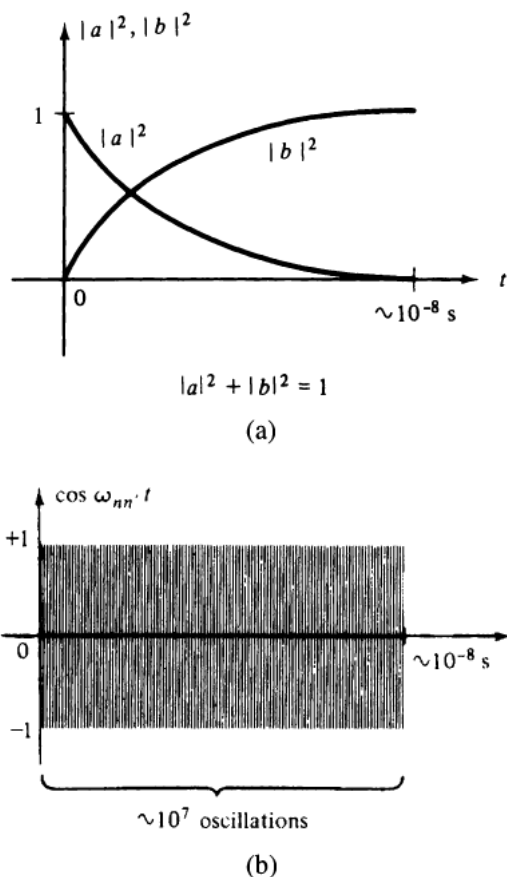


FIGURE 10.20 Radiation decay from the ψ_n state to the $\psi_{n'}$ state involves the superposition state $\psi = a\psi_n + b\psi_{n'}$ with time-varying coefficients a and b . The “beat” frequency $\omega_{nn'}$ between the ψ_n and $\psi_{n'}$ states is much greater than the switchover frequency of ψ .

where $\mathbf{d}$ is the dipole moment

$$\mathbf{d} = e\mathbf{r}_0, \quad \langle \mathbf{r} \rangle = \mathbf{r}_0 \cos \omega t \quad (10.138a)$$

We may apply this formula to calculate the power radiated when the hydrogen atom decays from the n th state to the (n') th state. From (10.137) we obtain

$$\mathbf{d} = 2e\mathbf{r}_{nn'} \quad (10.138b)$$

so that, with (10.138),

$$P = \frac{4}{3} \frac{(\omega_{nn'})^4 e^2}{c^3} |\mathbf{r}_{nn'}|^2 \quad (10.139)$$

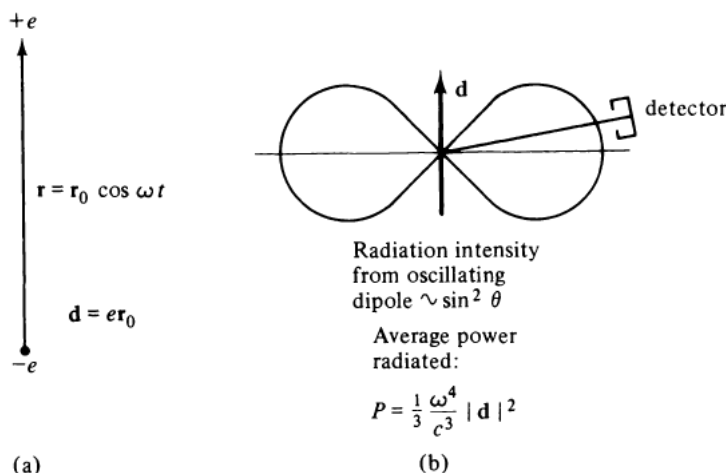


FIGURE 10.21 Energy characteristics of an oscillating dipole. (a) Dipole configuration. (b) Radiation profile of a dipole.

Calculation (see Problems 10.52 to 10.54) of the squared matrix element $|\mathbf{r}_{nn'}|^2$ with n standing for nlm and n' for $n'l'm'$ gives the following *selection rules*: The only conditions under which $|\mathbf{r}_{nn'}|^2$ (and therefore P) is not zero are

$$\Delta l = l' - l = \pm 1 \quad \text{and} \quad \Delta m_l = m' - m = 0, \pm 1 \quad (10.140)$$

For example, the transition $3S \rightarrow 1S$ ($\Delta l = 0$) is *forbidden*, as is the transition $3D \rightarrow 2S$ ($\Delta l = 2$). Such transitions are not accompanied by any (dipole) radiation and therefore are excluded by conservation of energy. The exclusion of the transitions between S states finds analogy with the classical theorem that spherically symmetric oscillatory charge distributions do not radiate.

The rule $\Delta l = \pm 1$, together with the law of conservation of angular momentum, indicates that for $\Delta l = -1$ the electromagnetic field (i.e., the photon) carries away angular momentum. As it turns out, photons have angular momentum quantum number equal to 1 and are therefore called *bosons*.

There are no restrictions on an atomic transition corresponding to change in the principal quantum number n . This is in agreement with the pre-Schrödinger spectral notation for emission from hydrogen: namely, the Lyman series corresponds to transitions from all n states to the ground state, the Balmer series corresponds to transitions to the $n = 2$ states, etc. (see Fig. 2.8).

PROBLEMS

10.51 What are the allowed transitions from the $5D$ states of hydrogen to lower states? Accompany your answer with a sketch representing these transitions.

- 10.52** With n representing the triplet nlm and n' the triplet $n'l'm'$, show that the matrix elements of $\mathbf{r}$ have the following complex representation:

$$|\mathbf{r}_{nn'}|^2 \equiv |\langle n|\mathbf{r}|n'\rangle|^2 = \frac{1}{2} \{ |\langle n|x+iy|n'\rangle|^2 + |\langle n|x-iy|n'\rangle|^2 \} + |\langle n|z|n'\rangle|^2$$

[Hint: Call $\langle n|x|n'\rangle \equiv x_{nn'}$, etc. Note also that $|\langle \mathbf{r} \rangle|$ denotes the magnitude of the vector $\langle \mathbf{r} \rangle$, while $|\langle x+iy \rangle|$ denotes the modulus of the complex variable $\langle x+iy \rangle$.]

- 10.53** Show that the matrix elements of $x \pm iy$ have the following integral representation:

$$(a) \quad \langle n|x \pm iy|n'\rangle = \int_{4\pi} [Y_l^m]^* Y_{l'}^{m'} \sin \theta e^{\pm i\phi} d\Omega \int_0^\infty R_{nl} R_{n'l'} r^3 dr$$

$$(b) \quad \langle n|z|n'\rangle = \int_{4\pi} [Y_l^m]^* Y_{l'}^{m'} \cos \theta d\Omega \int_0^\infty R_{nl} R_{n'l'} r^3 dr$$

- 10.54** Using the results of the last two problems and Tables 9.1 and 9.3, establish the selection rules for dipole radiation (10.140).
- 10.55** At the start of Section 10.7 it was noted that the energy eigenstates of an atom are stable against radiative decay, owing to their stationarity in time. However, these states do contain angular momentum, and one may argue that they therefore also contain rotating charge, which does radiate. Although the premise of this argument is correct, why is there still no radiation from the stationary states (from a classical point of view)?

Answer

An aggregate of N uniformly spaced point charges confined to move with fixed speed in a closed loop will radiate as a result of the acceleration of individual charges. However, in the limit that the charges approach a uniformly continuous distribution, $N \rightarrow \infty$, $\Delta \rightarrow 0$, $q \rightarrow 0$ (with total charge Nq and line charge density q/Δ constant), the radiation may be shown¹⁸ to vanish. This limiting case closely resembles the state of affairs for the stationary states of an atom. Although there is rotating charge, such charge is continuously distributed and, in accord with the classical prescription, does not radiate.

- 10.56** The interaction potential of an electron moving in the far field of a dipole $\mathbf{d}$ is

$$V = -\frac{ed}{r^2} \cos \theta$$

The dipole is at the origin and points in the z direction. The spherical coordinates of the electron are (r, θ, ϕ) .

- (a) Write down the time-independent Schrödinger equation for this system corresponding to zero total energy. What condition of the “atom” does this state describe?
- (b) Show that solutions to this equation are of the form $\varphi = r^s f(\theta, \phi)$. Obtain an equation for $f(\theta, \phi)$.
- 10.57** Two particles that are isolated from all other objects interact with each other through a central potential. As was established in Problem 10.23(d), the total angular mo-

¹⁸See Jackson, *Classical Electrodynamics*, Chapter 14.

mentum of the system may be written

$$\mathbf{L}_1 + \mathbf{L}_2 = \mathbf{L} + \mathbf{L}_{\text{CM}}$$

Show quantum mechanically that this total angular momentum is conserved.

10.58 (a) Prove that the *Runge-Lenz* vector

$$\hat{\mathbf{K}} = \frac{1}{2\mu e^2} [\mathbf{L} \times \hat{\mathbf{p}} - \hat{\mathbf{p}} \times \mathbf{L}] + \frac{\mathbf{r}}{r}$$

commutes with the Hamiltonian of the hydrogen atom (10.115).

(b) Show that the operator

$$\hat{\mathbf{A}} = \sqrt{-\mu e^4 / 2E} \hat{\mathbf{K}}$$

satisfies the commutation relations

$$[\hat{A}_x, \hat{A}_y] = i\hbar \hat{L}_z, \quad \text{etc.}$$

(c) Use the result above to show that the operators

$$\hat{\mathbf{B}}_+ = \frac{1}{2}(\hat{\mathbf{L}} + \hat{\mathbf{A}})$$

$$\hat{\mathbf{B}}_- = \frac{1}{2}(\hat{\mathbf{L}} - \hat{\mathbf{A}})$$

obey the angular momentum commutation relations and the equality $\hat{\mathbf{B}}_+^2 = \hat{\mathbf{B}}_-^2$.

(d) Derive the relation

$$\hat{\mathbf{B}}_+^2 + \hat{\mathbf{B}}_-^2 = \frac{1}{2} \left(\hbar^2 + \frac{\mu e^4}{2E} \right)$$

and use it to obtain the Bohr formula for the energy levels of hydrogen.

10.59 The classical harmonic oscillator with spring constant K and mass m oscillates at the *single* frequency (independent of energy)

$$\omega_0 = \sqrt{K/m}$$

The quantum mechanical oscillator, on the other hand, gives frequencies at all integral multiples of ω_0 , as follows directly from the eigenenergies

$$E_n = \hbar\omega_0 \left(n + \frac{1}{2} \right) \quad (n = 0, 1, 2, \dots)$$

If one end of the oscillator is charged, dipole radiation is emitted. In the classical domain, this radiation has frequency ω_0 . Show that selection rules that follow from calculation of the dipole matrix elements $x_{nn'}$ reduce the quantum mechanical spectrum to the classical one. (The concept of matrix elements of an observable is developed formally in Chapter 11.)

10.60 Consider a gas of noninteracting rigid dumbbell molecules with speeds small compared to the speed of light. The moment of inertia of each molecule is I .

- What is the Hamiltonian of a molecule in the gas?
- What are the eigenenergies of this Hamiltonian?
- Let a molecule undergo spontaneous decay between two rotational states. Owing to the recoil of the center of mass, there is a change in momentum of the center of mass of the molecule as well. Show that the frequency of the photon emitted in this process is

$$\nu = \nu_l \left(1 - \frac{\hbar \mathbf{k} \cdot \mathbf{n}}{Mc} \right)$$

The initial momentum of the center of mass is $\hbar \mathbf{k}$, $\mathbf{n}$ is a unit vector in the direction of the momentum of the emitted photon, and $\{\nu_l\}$ is the rotational line spectrum.

- What is the nature of the frequency spectrum emitted by the gas?

Answers

- Let $\mathcal{P}$ and M denote the momentum and mass, respectively, of the center of mass. Then

$$\hat{H} = \frac{\mathcal{P}^2}{2M} + \frac{\hat{L}^2}{2I}$$

$$(b) E_{k,l} = \frac{\hbar^2 k^2}{2M} + \frac{\hbar^2 l(l+1)}{2I}$$

- The frequency of photons emitted by a molecule is given by the change in energy

$$h\nu = \Delta E = \frac{\hbar^2 \mathbf{k} \cdot \Delta \mathbf{k}}{M} + h\nu_l$$

where $h\nu_l$ is written for the change in rotational energy. Since the molecule is a free particle, the momentum of the center of mass $\hbar \mathbf{k}$ can change only by virtue of the momentum carried away by the photon emitted in the transition. As a first approximation we will assume that the momentum carried away is $h\nu_l/c$. If $\hbar \mathbf{k}'$ is the momentum of the center of mass after emission, then by conservation of momentum we have (see Fig. 10.22)

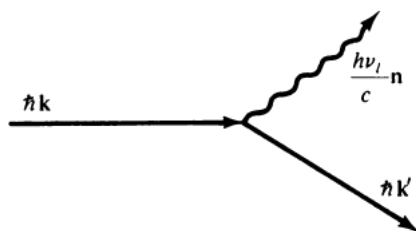


FIGURE 10.22 Change in linear momentum of rigid rotator due to recoil in the emission of a photon (see Problem 10.60).

$$\hbar \mathbf{k} = \hbar \mathbf{k}' + \frac{h\nu_l}{c} \mathbf{n}$$

so that

$$\hbar \Delta \mathbf{k} = \hbar(\mathbf{k}' - \mathbf{k}) = -\frac{h\nu_l}{c} \hat{\mathbf{n}}$$

Substituting this value in the expression above for $h\nu$ gives

$$\nu = \nu_l \left(1 - \frac{\hbar \mathbf{k} \cdot \mathbf{n}}{Mc} \right)$$

Since $\hbar k \ll Mc$, the assumption that momentum carried away by this radiation field is approximately $h\nu_l/c$ is justified.

- (d) The rotational spectrum $\{\nu_l\}$ remains a line spectrum with an infinitesimal broadening of lines.

10.61 Consider two identical rigid spheres of diameter a , which move in 3-space.

- (a) What is the Hamiltonian of this system?
- (b) Separate out the center-of-mass motion to obtain an equation for the wavefunction for relative motion, $\varphi(\mathbf{r})$.
- (c) What boundary conditions must $\varphi(\mathbf{r})$ satisfy?
- (d) What are the eigenstates and eigenenergies for this system?
- (e) What are the parities of these eigenstates?
- (f) What is the radial probability, $P(r)$, for this system?

Answers (partial)

$$(a) \hat{H}(1, 2) = \frac{\hat{p}_1^2}{2m} + \frac{\hat{p}_2^2}{2m} + V(|\mathbf{r}_1 - \mathbf{r}_2|)$$

$$V(|\mathbf{r}_1 - \mathbf{r}_2|) \equiv V(r) = \infty, \quad r \leq a$$

$$V(r) = 0, \quad r > a$$

$$(b) \left[\frac{\hat{p}_r^2}{2\mu} + \frac{\hat{L}^2}{2\mu r^2} + V(r) - E \right] \varphi(\mathbf{r}) = 0$$

$$(c) \varphi_{klm}(\mathbf{r}) = 0, \quad r \leq a$$

$$(d) \varphi_{klm} = \varphi_{lm}(kr) Y_l^m(\theta, \phi), \quad r > a$$

$$\varphi_{lm}(kr) = A[n_l(ka)j_l(kr) - j_l(ka)n_l(kr)]$$

$$E_k = \frac{\hbar^2 k^2}{2\mu}$$

- (e) Referring to Problem 9.46, we obtain

$$\hat{\mathbb{P}}\varphi_{lm}(kr) = (-)^l \varphi_{lm}(kr)$$

10.8 ■ THOMAS-FERMI MODEL

The Thomas–Fermi model is appropriate to atoms with sufficient number of electrons which permits the system to be treated in a statistical sense. The model views an atom as a spherically symmetric gas of Z electrons which surrounds a nucleus of charge Ze . It is further assumed that the potential in the medium, $\Phi(r)$, varies sufficiently slowly so that it may be taken as constant in small volumes of the electron gas. The description of the Fermi energy, such as given in Section 2.3, then applies. (See Fig. 10.23.) There, we recall the Fermi energy was equated to the maximum kinetic energy of an electron in an aggregate of electrons at 0 K. This description maintains in the present model since at normal temperature $k_B T \ll \Phi(r)$. (The Fermi energy is discussed further in Section 12.9.)

The relation between E_F and electron density $n(r)$ is given in Problem 2.42, and with the preceding description we write

$$E_{\text{max}}^{\text{Kin}} = E_F = \frac{\hbar^2}{2m} [3\pi^2 n(r)]^{2/3} \quad (10.141)$$

The energy of an electron in the medium at a distance r from the nucleus is

$$E(r) = \frac{p^2}{2m} - e\Phi(r) \quad (10.142)$$

where we have written $e = |e|$. The preceding equation indicates that for $p^2/2m > e\Phi(r)$, $E > 0$, which implies that the electron escapes the atom. As this does not occur in the present model, we may conclude that maximum

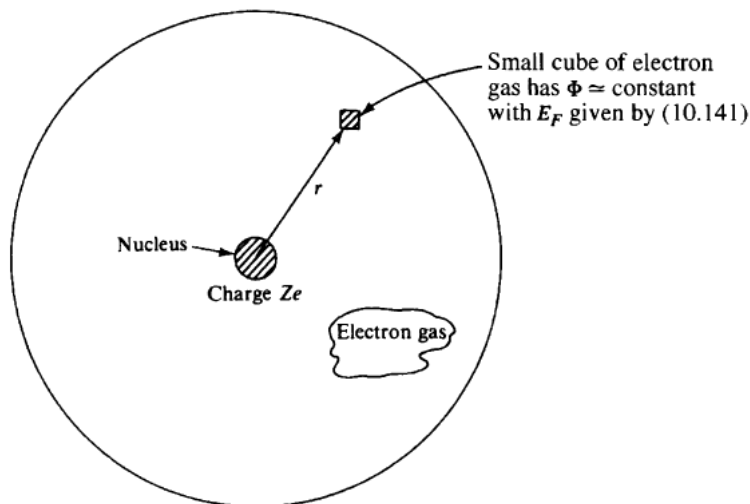


FIGURE 10.23 The Thomas–Fermi statistical atom.

kinetic energy may also be set equal to $e\Phi$. With (10.141) we may then write

$$e\Phi(r) = \frac{\hbar^2}{2m} [3\pi^2 n(r)]^{2/3}$$

An additional relation between the potential $\Phi(r)$ and density $n(r)$ is given by Poisson's equation (cgs),

$$\nabla^2 \Phi = 4\pi en$$

Working in the given spherically symmetric geometry, and eliminating $n(r)$ from the latter two equations, gives

$$\nabla^2 \Phi = \frac{1}{r} \frac{d^2}{dr^2} (r\Phi) = \frac{4\pi e}{3\pi^2} \frac{(2me)^{3/2}}{\hbar^3} \Phi^{3/2} \quad (10.143)$$

This is the desired single equation for $\Phi(r)$.

The following boundary conditions apply to the potential. Near the origin $\Phi(r)$ is due predominantly to the nucleus, and we write

$$\Phi \simeq \frac{Ze}{r}$$

or, equivalently,

$$\lim_{r \rightarrow 0} r\Phi(r) = Ze \quad (10.144a)$$

At large r we obtain

$$\lim_{r \rightarrow \infty} r\Phi(r) = 0 \quad (10.144b)$$

The resulting equation (10.143) may be written in a more concise form in terms of new dimensionless variables (x, χ) defined by

$$r = ax \quad (10.145a)$$

$$a \equiv \frac{1}{2} \left(\frac{3\pi}{4} \right)^{2/3} \frac{\hbar^2 / me^2}{Z^{1/3}} = \frac{0.885a_0}{Z^{1/3}} \quad (10.145b)$$

$$\Phi(r) \equiv \frac{Ze}{r} \chi(x) \quad (10.145c)$$

When written in terms of these new variables (see Problem 10.68), (10.143) reduces to

$$\boxed{x^{1/2} \frac{d^2 \chi}{dx^2} = \chi^{3/2}} \quad (10.146)$$

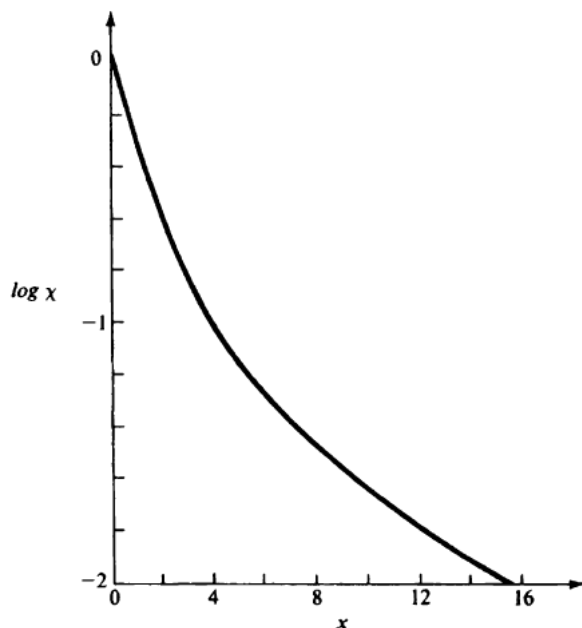


FIGURE 10.24 Semilogarithmic plot of solution to the Thomas–Fermi equation (10.146) subject to the boundary conditions (10.147).

This is the standard form of the (highly nonlinear) Thomas–Fermi equation. Boundary conditions (10.144), when written in terms of the new potential $\chi(x)$, appear as

$$\lim_{x \rightarrow 0} \chi = 1, \quad \lim_{x \rightarrow \infty} \chi = 0 \quad (10.147)$$

The semilogarithmic plot shown in Fig. 10.24 reveals the rapid decay of the potential χ with increase in x . Thus, for example, for $x > 4$, $\chi < 0.1$.¹⁹

Variation of Atomic Size with Z

An important conclusion of the preceding model is as follows: It is evident from (10.145b) that the parameter a may be viewed as an effective radius of the atom (i.e., the radius of a sphere which contains a fixed fraction of atomic electrons). Thus there is an implied decrease in effective atomic size with increase in Z . However, it should be borne in mind that the shell structure of atoms is lost in the Thomas–Fermi model. Thus one might expect the implied decrease in atomic size to be roughly valid within a given atomic shell of fixed principal quantum number. Observations find this property to be approximately obeyed. (See Fig. 12.11b.) The closely allied “Hartree–Fock” approximation is described in Section 13.10.

¹⁹Numerical integration of the Thomas–Fermi equation (10.146) was originally performed by V. Bush and S. H. Caldwell, *Phys. Rev.* **38**, 1898 (1931).

PROBLEMS

- 10.62** A hydrogen atom suffers a decay from the $3D$ to the $2P$ state and emits a photon.
- What is the frequency of this photon (in Hz)?
 - Given that the decay time of this transition is $\simeq 10^{-10}$ s, what is the power (in W) emitted in the transition?
- 10.63** Show that the three φ_{2P} states (10.134) comprise an orthonormal set. What is the value of the constant A ? Leave your answer in terms of square roots and a_0 .
- 10.64** Of what angular momentum operators are the respective three φ_{2P} states (10.134) eigenfunctions?

Answer

As described in Chapter 9, when operating a wavefunction $f(\mathbf{r})$, the operators $\hat{L}_x$, $\hat{L}_y$, $\hat{L}_z$ effect a rotation of $\mathbf{r}$ about the x , y , z axes, respectively. Thus φ_{2P_x} is an eigenstate of $\hat{L}_x$; φ_{2P_y} is an eigenstate of $\hat{L}_y$; and φ_{2P_z} is an eigenstate of $\hat{L}_z$. All three functions are eigenstates of $\hat{L}^2$.

- 10.65** Show that $\langle \psi_1 | \psi_2 \rangle = 0$ for the orbitals (10.135).
- 10.66** (a) Show that the function φ_{2P_x} as given by (10.134a) is maximum along the x axis.
- (b) With this property at hand, argue that ψ_1 as given by (10.135a) exhibits maxima along the $(1, 1, 1)$ directions.

Answers (partial)

- (a) We may write

$$\varphi_{2P_x} = A \sin \theta \cos \phi r e^{-r/2a_0}$$

A polar plot of $\sin \theta$ reveals a circle tangent to the z axis at the origin. Thus $\sin \theta$ generates a circular torus about the z axis. A polar plot of $\cos \phi$ reveals a circle in the xy plane tangent to the y axis at the origin and with maximum at $\phi = 0$ (positive x axis). The intersection of these two factors, $\sin \theta \cos \phi$, is evidently maximum along the positive x axis. The remaining r -dependent component merely modulates this result.

- 10.67** This problem concerns annihilation and creation operators for spherical Bessel functions.²⁰ Introducing the operators (for $l > 0$)

$$b_l^\pm = -\frac{i}{x} \frac{d}{dx} x \pm i \frac{l}{x}$$

- (a) Show that

$$\hat{b}_l^+ \hat{b}_l^- = \hat{b}_{l+1}^- \hat{b}_{l+1}^+ = -\frac{1}{x} \frac{d^2}{dx^2} + \frac{l(l+1)}{x^2}$$

- (b) Show that Bessel's equation (10.54) may be written

$$(\hat{b}_l^+ \hat{b}_l^- - 1) R_l(x) = 0$$

²⁰L. Infeld, *Phys. Rev.* **59**, 737 (1941).

(c) Show that

$$\hat{b}_{l+1}^+ R_l = R_{l+1}$$

(d) Show that

$$\hat{b}_l^- R_l = R_{l-1}$$

(e) Apart from normalization, employ the preceding formalism to obtain:

i. $j_1(x)$ from $j_0(x)$.

ii. $n_1(x)$ from $n_0(x)$

10.68 The radial wavefunction satisfies the second-order differential equation (10.109). The student may recall that any such equation has two independent solutions. Show that for bound states there is only one linearly independent solution for given l and E values.

Answer

The equation may be written

$$\left[\hat{D}^2 - \frac{l(l+1)}{r^2} + \frac{2\mu}{\hbar^2}(E - V) \right] u = 0$$

$$\hat{D} \equiv \frac{d}{dr}$$

$$u \equiv rR$$

Let solutions to the preceding equation be written u_1 and u_2 . Multiplying the equation for u_2 by u_1 and vice versa and subtracting the resulting equations gives

$$u_1 \hat{D}^2 u_2 - u_2 \hat{D}^2 u_1 = 0$$

or, equivalently,

$$\hat{D}[u_1 \hat{D} u_2 - u_2 \hat{D} u_1] = 0$$

Integrating, we find

$$u_1 \hat{D} u_2 - u_2 \hat{D} u_1 = C$$

where C is a constant. For bound systems one may assume that either $u_1, u_2 \rightarrow 0$ or $\hat{D} u_1, \hat{D} u_2 \rightarrow 0$ as $r \rightarrow \infty$, which, in either case, gives $C = 0$. There results

$$\frac{d \ln u_2}{dr} = \frac{d \ln u_1}{dr}$$

which gives

$$u_2 = a u_1$$

where a is a constant. Thus, u_2 and u_1 are not independent.

10.69 Employing the transformation equations (10.145), show that (10.143) reduces to the Thomas-Fermi equation (10.146).

10.70 This problem addresses motion of an electron in a steady magnetic field as described in Section 10.4.

- (a) Find the mean square radius a^2 of an electron moving in a plane normal to a steady magnetic field.
- (b) Obtain your answer to part (a) once again, but now through use of the correspondence principle.

Answers

- (a) Referring to (10.86) et seq. indicates that

$$a^2 = \langle (y - y_0)^2 \rangle$$

Due to the harmonic oscillator structure of (10.88), we may write

$$\langle (y - y_0)^2 \rangle = \frac{2}{K} \langle V \rangle$$

where V is potential energy. With Problem 7.10 we may then write

$$a^2 = \frac{2}{K} \times \frac{\langle E \rangle}{2} = \frac{1}{K} \langle E \rangle$$

$$a^2 = \frac{\hbar}{m\Omega} \left(n + \frac{1}{2} \right)$$

- (b) The classical kinetic energy of the electron is

$$\frac{1}{2}mv^2 = \frac{1}{2}ma^2\Omega^2 = \frac{1}{2}E$$

Thus

$$a^2 = \frac{E}{m\Omega^2}$$

Inserting the quantum value for E gives the desired result.

- 10.71** You are asked to design an absorbing sink for rotationally polarized radiation of frequency 2.5×10^{-10} Hz. You propose to use a gas of dipole molecules. What is the moment of inertia, I (cgs), of molecules that will perform this task in $l = 0 \rightarrow l = 1$ transitions? What is the effective radius, r_0 (in Å), of these molecules if their reduced mass is 30 amu? (See Problem 1.23.)
- 10.72** (a) Show that $\varphi_{lm}(r, \theta, \phi) = r^l Y_l^m(\theta, \phi)$ are eigenfunctions of the free-particle Hamiltonian. What are the corresponding eigenenergies of these states?
- (b) Show that this finding is consistent with the free-particle eigenproperties (10.56).
- (c) What is the order of degeneracy of these eigenenergies? [Hint: For part (b), consult Table 10.1.]
- 10.73** (In the following, hats over operators are deleted.) The formulation for evaluating the momentum, p_n , conjugate to a given coordinate, q_n , for a system described

by N curvilinear coordinates is based on the following three properties:²¹ (i) the differential volume element: $dV = D dq_1 \dots dq_N$; (ii) the commutator: $[q_n, p_n] = i\hbar$; (iii) Hermiticity: $\langle \phi | p_n \psi \rangle = \langle p_n \phi | \psi \rangle$. The momentum operator conjugate to q_n is then given by

$$p_n = -i\hbar \frac{1}{\sqrt{D}} \frac{d\sqrt{D}}{dq_n}$$

- Obtain an expression for the radial momentum operator, p_ρ , appropriate to cylindrical coordinates (Fig. 1.7).
- Show that for the operator you have found, $[\rho, p_\rho] = i\hbar$.
- Show that the operator p_ρ is Hermitian (with respect to square integrable functions in a plane of constant z).
- Show that this operator obeys the following relation. (An expression for ∇_ρ^2 is given in Appendix D.)

$$p_\rho^2 = -\hbar^2 \left[\nabla_\rho^2 - \frac{1}{4\rho^2} \right]$$

- With your answer to part (d), write down a “corrected” Hamiltonian in cylindrical coordinates for a particle moving in the presence of a potential field, $V(\rho, \phi)$.
- Show that your answer to part (e) gives the correct Schrödinger equation for a free particle in cylindrical coordinates (10.75a).
- Show that your answer to part (e) reduces to the classical Hamiltonian in the classical limit.

Answers (partial)

- $p_\rho = -i\hbar \frac{1}{\rho^{1/2}} \frac{d\rho^{1/2}}{d\rho} = -i\hbar \left(\frac{1}{2\rho} + \frac{d}{d\rho} \right)$
- $H = \frac{1}{2m} \left[p_\rho^2 + \frac{p_\phi^2 - \hbar^2(1/2)^2}{\rho^2} \right] + V(\rho, \phi)$
- Setting $\hbar = 0$ returns the classical Hamiltonian.

10.74 What are the eigenfunctions, good quantum numbers, and eigenenergy degeneracies for the following configurations?

- A free particle moving in three dimensions in (i) Cartesian, (ii) spherical, and (iii) cylindrical coordinates.
- A particle confined to a (i) cubical box, (ii) spherical box, and (iii) cylindrical box.
- In each case, state which type of degeneracy is described. (See Section 8.5.) For infinite degeneracies discuss the nature of these infinities (i.e., continuous or discrete).

²¹W. Pauli, “Die Allgemeinen Prinzipien der Wellenmechanik,” *Hand. d. Physik* **XXIV**, 83–272 (1933).

- 10.75** An electron is free to move in the domain exterior to a rigid, impenetrable, smooth-walled, charge-neutral sphere of radius a .
- (a) What are the eigenfunctions and eigenenergies of this system?
 - (b) Write down the normalization condition for these wavefunctions. (*Hint*: See Problem 10.60.)
- 10.76** An electron is confined to the interior domain of a cylindrical cavity of radius a and length L . Photons at 3.0 eV are incident on the cavity and raise the electron from the ground state to the first excited state. What is the radius a (in Å) of the cylinder, given that $L = 4.7$ Å?
- 10.77** What is the value of $\left\{ \exp i2\pi \left[\sin \left(\frac{\pi/2}{l(l+1)} \frac{\hat{L}^2}{\hbar^2} \right) \right] \right\} Y_{l,m}$?